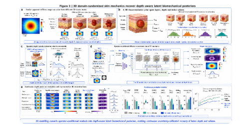
2026-05-31 10:11:05 You Asked

分支 · Computatio***electronics.pdf
4.4 MB



对于FIgure 3的数据，结合我们之前的讨论的设计（其他的细节可以参考我们的聊天记录，还需要考虑到深度信息的多峰特征）（以及ultrasound和piezoelectric sheet的benchmark，尤其是后者，在深度信息的解析方面应该有天然的劣势），如何让Codex先检查涉及到的数据是否存在，然后根据设计图和PDF里面对于数据的规划，补充缺失的数据，然后把数据合成到Figure3，再用API的Image 2.0和NanoBanana做出最高美感的图片。同时注意要有衔接第四张图（posterior合成）的相关部分

ChatGPT gpt-5-5-pro

我的建议是把 Codex 任务拆成 **data audit → data synthesis/completion → Figure 3 deterministic rendering → Image API/NanoBanana aesthetic pass → locked-data compositing**。核心原则：**Figure 3 的科学信息必须由代码生成，Image 2.0/NanoBanana 只做视觉润色，不能改动数值图、曲线、坐标轴、posterior shape 和 panel labels。**

你的 Figure 3 现在应该被定义为：从 Figure 2 的 $E_{\mathrm{app}}(x,y)$ 升级到 $E(x,y,z)$，重点不是跨硬件大比拼，而是 **3D latent mechanics + depth-aware posterior**；Figure 4 再接 reliability、experimental anchoring、uncertainty calibration、OOD/failure detection 和 task-conditioned utility。这个逻辑和之前 Figure 2–4 的设计一致：Figure 3 负责说明 2D apparent stiffness 不够，并进入 continuous depth posterior；Figure 4 才承接 posterior synthesis 和 reliability analysis。

---

1. 给 Codex 的总体任务定义

让 Codex 先不要直接“画图”。第一步必须生成一个 `Figure3_data_manifest.json`，逐项检查数据是否存在，并明确每个 panel 的数据来源：

```text
REAL_EXPERIMENTAL    = 已真实测量的数据，例如 MMP phantom / animal / ex vivo data
SIMULATED_BENCHMARK  = 根据同一 latent E(x,y,z) 通过 operator forward model 合成
SYNTHETIC_PLACEHOLDER = 只用于概念图或临时排版，不用于主文定量 claim
MISSING              = 缺失，不能伪装成真实结果
```

这点很关键。MMP 可以有实验 anchor，因为 Research Strategy 里已有 MMP 与 mechanical tester / standard elastomer 的 calibration、cylindrical inclusion、corner-stiffness、stiffness-gradient phantom，以及 Young's modulus spatial mapping 的 preliminary foundation。 但 ultrasound 和 piezoelectric sheet 在 Figure 3 里应主要定位为 **simulation-derived benchmark operators**，不能画成"三台真实设备测了同一个病灶"的真实 head-to-head experiment。

---

2. Figure 3 必须检查的 data contract

让 Codex 建一个 `configs/figure3_required_assets.yaml`，大致如下：

```yaml
figure3:
  global:
    required:
      - data/fig3/latent_truth/volume_E_xyz.npz
      - data/fig3/metadata/cases.csv
      - data/fig3/metadata/operator_metadata.csv
      - data/fig3/metadata/train_depth_support.json
    optional:
      - data/fig2/eapp_outputs/eapp_cases.npz
      - data/fig4/posterior_bank/posterior_samples_fig3.npz

  panel_a_nonidentifiability:
    required:
      - data/fig3/panel_a/eapp_common.npy
      - data/fig3/panel_a/alternative_3d_states.npz
      - data/fig3/panel_a/projections.npy

  panel_b_domain_randomization:
    required:
      - data/fig3/domain_randomization/parameter_table.csv
      - data/fig3/domain_randomization/example_volumes.npz
      - data/fig3/domain_randomization/histograms.csv

  panel_c_operator_kernels:
    required:
      - data/fig3/operators/mmp_kernel.npz
      - data/fig3/operators/piezo_sheet_kernel.npz
      - data/fig3/operators/ultrasound_strain_kernel.npz
      - data/fig3/operators/lateral_psf.csv
      - data/fig3/operators/operator_readouts_examples.npz

  panel_d_model_schematic:
    required:
      - data/fig3/model_io/io_shapes.json
```

```yaml
      - data/fig3/model_io/posterior_outputs.json

  panel_e_reconstruction_gallery:
    required:
      - data/fig3/posteriors/posterior_samples_E_xyz.npz
      - data/fig3/posteriors/posterior_mean_volume.npz
      - data/fig3/posteriors/posterior_uncertainty_volume.npz
      - data/fig3/posteriors/zbottom_samples.npz
      - data/fig3/posteriors/forward_residuals.csv

  panel_f_metrics:
    required:
      - data/fig3/metrics/depth_metrics.csv
      - data/fig3/metrics/calibration_metrics.csv
      - data/fig3/metrics/risk_coverage.csv
      - data/fig3/metrics/error_vs_extrapolation.csv
      - data/fig3/metrics/multimodality_metrics.csv

  fig4_bridge:
    required:
      - data/shared_posterior/fig3_to_fig4_posterior_bank.npz
      - data/shared_posterior/fig3_to_fig4_manifest.json
```

Codex 的第一条硬性规则：**缺任何 required asset，先 report，不画最终图；只有 simulation benchmark 缺失时才允许按设计补齐。真实实验数据缺失时只能标记为 MISSING 或 placeholder，不得自动编造。**

---

3. 缺失数据应该怎么补

3.1 latent 3D skin mechanics library

Codex 应补齐的是 simulation library，而不是"伪真实实验"。生成参数包括：

```text

Layer:
  epidermis_thickness
  dermis_thickness
  subcutis_thickness
  layer_moduli

Lesion:
  z_top
  z_bottom
  thickness = z_bottom - z_top
  radius
  stiffness_contrast C
```

```
  rim_stiffness
  heterogeneous_core
  boundary_irregularity
  multifocality

Operator / acquisition:
  contact_pressure
  lateral_psf
  sensor_pitch
  readout_noise
  calibration_shift
  depth_kernel Kd(z)
```

这与 Figure 3b 的设计一致：3D domain-randomized skin mechanics 要覆盖 layer modulus、lesion depth、thickness、contrast、boundary irregularity、heterogeneous core、contact state 和 operator noise。

3.2 MMP、piezo sheet、ultrasound 的 forward operators

三者必须共享同一个 latent truth $E(x, y, z)$，但保留 native readout：

```text
MMP:
  y_mmp = H_mmp(E, q_mmp) + noise
  readout type: Hall / magnetic coupling matrix
  depth kernel: near-to-intermediate depth sensitivity
  lateral behavior: multi-point indentation + lateral coupling

Piezoelectric sheet:
  y_piezo = H_piezo(E, q_piezo) + noise
  readout type: surface voltage map
  depth kernel: strongly surface-biased exponential decay
  expected weakness: poor z_bottom observability, broad or multimodal depth posterior

Ultrasound strain:
  y_us = H_us(E, q_us) + noise
  readout type: strain field / elastographic surrogate
  depth kernel: broader and deeper field
  expected tradeoff: better depth reach than piezo sheet, but more lateral blur / model-
assumption dependence
```

这要体现在 Figure 3c：**operator depth kernels determine what is recoverable**。MMP 应显示 near-to-intermediate sensitivity；piezo sheet 显示 surface-biased kernel；ultrasound 显示 broader/deeper field。重点不是硬件排名，而是 depth observability。

3.3 piezoelectric sheet 的“天然劣势”如何进入数据

不要只在文字里说 piezo 差。Codex 应让这个劣势体现在三个层面：

```text
1. Kernel:
   K_piezo(z) = exp(-z / lambda_piezo), lambda_piezo small, e.g. 0.3-0.8 mm.

2. Posterior:
   deep lesion cases 的 p(z_bottom | y_piezo) 应该更宽、可能多峰；
   low-contrast + deep cases 不能给出过度自信的单峰。

3. Metrics:
   piezo sheet 在 depth MAE / CRPS / Width90 / AUROC_fail / error-vs-d_out 上劣于 MMP 和
ultrasound；
   但在 near-surface apparent stiffness map 上可以表现合理，避免把它画成“全面失败”。
```

这符合你之前的 benchmark 设定：Figure 3 的 claim 不应是“ours 的 3D volume 最清楚”，而应是 **operator-conditioned posterior sampling improves continuous depth posterior, coverage-width tradeoff, and failure detection under unseen-depth extrapolation**。

---

4. 多峰 depth posterior 必须显式建模

Figure 3 不能只画一个高斯样的 $p(z_{\mathrm{bottom}})$。你的设定里，深度信息可能多峰，尤其是：

```text
shallow low-contrast lesion
deep high-contrast lesion
thick intermediate lesion
layered-background artifact
boundary/contact-induced spreading
```

这些不同 3D tissue states 可以给出相似 $E_{\mathrm{app}}(x, y)$，所以 $p(z_{\mathrm{bottom}})$ 应该允许 bimodal / multimodal。Figure 3 的反问题比 Figure 2 更 ill-posed：深度变量弱可观测，2D apparent map 对 3D volume 是 many-to-one，posterior 可能宽甚至多峰。

让 Codex 在 `metrics/multimodality_metrics.csv` 中保存：

```csv
case_id,operator,method,n_modes,mode_1_mm,mode_2_mm,mode_1_weight,mode_2_weight,valley_ratio,hdi90_components,multimodal_flag
```

用 KDE 或 Gaussian mixture 检测 posterior modes：

```python
```

```python
def detect_depth_modes(samples):
    # samples: posterior samples of z_bottom, shape [M]
    # output: mode positions, mode weights, multimodality score
    kde = fit_1d_kde(samples)
    grid = np.linspace(0, 15, 600)
    density = kde(grid)
    peaks = find_local_peaks(density, min_prominence=0.08 * density.max())
    modes = grid[peaks]
    weights = integrate_density_around_modes(grid, density, peaks)
    valley_ratio = compute_valley_ratio(grid, density, peaks)
    return modes, weights, valley_ratio
```

Figure 3e/f 里至少放 3 条 posterior curves：

```text
MMP shallow case: narrow unimodal posterior
MMP low-contrast/deep case: broad or bimodal posterior
Piezo deep case: broad / multimodal / high uncertainty
Ultrasound deep case: deeper posterior but lateral uncertainty tradeoff
```

主图不要再放 band accuracy。评价单位应是每个样本的 continuous depth posterior $F_i(z_{\text{bottom}})$，band 头只作为 ordinal auxiliary task；顶层汇报也应先 config-level 聚合，再对 held-out depth group 做 macro average，避免重复 scan 虚高样本数。

---

5. Figure 3f 的统计图应该固定这些指标

不要用表格。用小型 bar plots、ridge plots、calibration bars、risk–coverage curve 和 error-vs-extrapolation scatter。

```text
Continuous error:
  MAE(z_bottom)
  RMSE(z_bottom)
  Bias(z_bottom)
  P90AE(z_bottom)

Probabilistic posterior:
  CRPS(z_bottom)

Calibration:
  Cov90(z_bottom)
  Width90(z_bottom)

Interval geometry:
```

```
  IoU([z_top, z_bottom])

Failure / OOD:
  AUROC_fail using total_std / epi_std / entropy
  Risk-coverage / AUSE
  error vs d_out
  extrapolation slope β
  range compression ratio
```

这些指标比 band accuracy 更合适，因为 band accuracy 会把 0.1 mm 的相邻错误和 2 mm 的远距离错误都记为同类错误。你之前的 depth evaluation note 已经明确建议用 MAE、RMSE、Bias、P90AE、CRPS、coverage、width、interval IoU、risk-coverage 和 OOD failure detection。

---

6. Figure 3 和 Figure 4 的衔接

Figure 3 不应在最后"结束"。它应该输出一个 **posterior bank**，供 Figure 4 做 posterior synthesis / uncertainty calibration / failure detection。

让 Codex 在 Figure 3 生成结束时保存：

```text
data/shared_posterior/fig3_to_fig4_posterior_bank.npz
  posterior_samples_zbottom
  posterior_samples_ztop
  posterior_samples_thickness
  posterior_samples_E_xyz
  posterior_mean_E_xyz
  posterior_std_E_xyz
  operator_id
  case_id
  method_id
  depth_group
  train_depth_support
  uncertainty_scores
  forward_residual
  failure_label_tau_1mm
  multimodality_metrics

data/shared_posterior/fig3_to_fig4_manifest.json
  panel_e_cases_used
  panel_f_metrics_used
  paths_for_fig4_reliability
  paths_for_fig4_task_utility
```

Figure 3d 或 Figure 3f 的右下角加一个小的 connector：

```text
Posterior bank
p(E(x,y,z), ztop, zbottom | yd, d, qd)
→ Fig. 4 reliability, calibration and task utility
```

这样 Figure 4 可以自然接：MMP experimental anchoring、reliability curve、PIT、risk–coverage、AUROC/AUPRC、task utility heatmap。之前 Figure 4 的设计已经把 experimental anchoring、uncertainty calibration/failure detection 和 operator utility 作为主轴。

---

7. 直接给 Codex 的 prompt

下面这段可以直接放到 Codex 里。

```text
You are working on Figure 3 for a Nature Computational Science-style manuscript on operator-conditioned inverse biomechanical sensing.

Goal:
Build a fully data-driven Figure 3: "3D domain-randomized skin mechanics recover depth-aware latent biomechanical posteriors".
Do NOT fabricate real experimental data. Missing real experimental data must be marked MISSING.
Missing simulation-derived benchmark data may be generated from the specified simulator with deterministic seeds and clearly labeled SIMULATED_BENCHMARK.

Scientific framing:
Figure 3 must transition from 2D apparent stiffness E_app(x,y) to latent 3D tissue mechanics E(x,y,z). It must show why 2D apparent maps are non-identifiable, how layered 3D skin mechanics are domain-randomized, how operator depth kernels determine recoverable information, and how operator-conditioned diffusion returns continuous, potentially multimodal depth posteriors.
Figure 4 will reuse Figure 3's posterior bank for reliability, calibration, OOD/failure detection and task-conditioned utility.

Hard constraints:
1. First run a data inventory. Create reports/figure3_data_manifest.md and data/fig3/Figure3_data_manifest.json.
2. Every panel asset must be labeled REAL_EXPERIMENTAL, SIMULATED_BENCHMARK, SYNTHETIC_PLACEHOLDER or MISSING.
3. Real MMP phantom/animal/human data must never be synthesized. If absent, mark missing and use simulation-only Figure 3 panels.
4. Ultrasound and piezoelectric sheet are simulation-derived benchmark operators unless real paired data exist.
5. Piezoelectric sheet must be represented as surface-biased and naturally weak for depth recovery.
6. Depth posterior must allow multimodality. Do not collapse posterior to mean ± std only.
7. Figure 3f must not use tables. Use plots only.
```

```
8. Final Image API/NanoBanana steps may enhance aesthetics but must not alter data plots, axes,
numeric values, posterior curves, text labels, or panel letters.

Required scripts:
- scripts/fig3_inventory.py
- scripts/fig3_generate_missing_simulation.py
- scripts/fig3_compute_posteriors_and_metrics.py
- scripts/fig3_render_vector.py
- scripts/fig3_ai_refine_locked_layers.py
- scripts/fig3_composite_locked_data.py

Required output:
- figures/figure3/figure3_data_driven.svg
- figures/figure3/figure3_data_driven.pdf
- figures/figure3/figure3_data_driven.png, 600 dpi
- figures/figure3/figure3_ai_refined.png
- figures/figure3/figure3_final_locked_overlay.png
- data/shared_posterior/fig3_to_fig4_posterior_bank.npz
- data/shared_posterior/fig3_to_fig4_manifest.json
- reports/figure3_data_manifest.md
- reports/figure3_methods_data_provenance.md

Panel specifications:
A. Non-identifiability:
Show one common E_app(x,y) and 4 different 3D states producing similar apparent map:
shallow thin lesion, deep high-contrast lesion, thick intermediate lesion, heterogeneous core
lesion.

B. 3D domain-randomized skin mechanics:
Show layered skin block with epidermis, dermis, subcutis, deeper tissue.
Show parameter distributions for z_bottom, lesion thickness, contrast C, layer modulus
variation, contact/noise/calibration.

C. Operator depth kernels:
Show MMP, piezoelectric sheet and ultrasound strain array.
Plot K_d(z) and lateral h_d(r).
MMP: near-to-intermediate depth sensitivity.
Piezo sheet: surface-biased, steep decay with depth, weak z_bottom observability.
Ultrasound: broader/deeper field, lower lateral specificity.
Do not frame this as hardware ranking.

D. 3D operator-conditioned diffusion:
Input: yd, operator token d, metadata qd, optional E_app/M.
Output: posterior samples of E(x,y,z), z_top, z_bottom, thickness, uncertainty volume, forward
residual.
Add connector: Posterior bank → Figure 4 reliability/calibration/task utility.

E. Representative 3D posterior gallery:
```

```
Cases: shallow lesion, mid-depth lesion, deep lesion, irregular boundary, low contrast /
ambiguous, OOD depth.
Columns: readout, true 3D slice, posterior mean, uncertainty volume, E_app projection,
p(z_bottom), forward residual.
For ambiguous/deep cases, draw multimodal posterior curves when supported by samples.

F. Continuous posterior metrics:
No table. Include:
MAE, RMSE, Bias, P90AE for z_bottom;
CRPS;
Cov90 + Width90;
interval IoU for [z_top,z_bottom];
risk-coverage/AUSE;
error vs extrapolation distance d_out;
multimodality frequency by operator.
Compare at least:
Direct scalar regression,
2.5D/3D U-Net,
Kernel/GP depth-response baseline,
Vanilla 3D diffusion,
Mask-only or E_app-only conditional diffusion,
DPS-style inverse diffusion,
Operator-conditioned 3D diffusion (ours).
Also include operator comparison MMP vs piezo sheet vs ultrasound where relevant, with same
latent truth and matched-SNR setting.

Implementation details:
- Use deterministic random seed 20260531.
- Store all synthetic data and metrics as CSV/NPZ before plotting.
- Use shared latent truth across operators.
- Preserve native readout types; do not force all operators into the same image format.
- Use matched-SNR benchmark in main Figure 3 and optionally realistic-noise benchmark in
supplement.
- For multimodal posteriors, compute KDE/GMM-based n_modes, mode locations, mode weights and
HDI90 components.
- For Figure 4 bridge, save posterior samples, uncertainty scores, failure labels and risk-
coverage inputs.

Rendering:
- Use vector graphics for all plots and labels.
- Use consistent panel letters a-f.
- Use clean white background, thin blue section rules, compact Nature-style typography.
- Use colorbars only where needed and keep units: kPa, mm.
- Use no table in the final image.
```

---

8. Image 2.0 / NanoBanana 的正确用法

OpenAI 当前图像文档说明，Images API 或 Responses API 可用于生成/编辑图像，`gpt-image-2` 支持根据文本和图像输入进行图像生成或编辑。(OpenAI платформы) 这里建议把它当成 **visual polish model**，不是数据绘图工具。也就是说：

```text
Step 1: Codex / matplotlib / SVG 生成真实 data-bearing figure scaffold.
Step 2: 导出 data-locked raster + semantic masks。
Step 3: Image 2.0 只润色 schematic panels，例如 3D tissue blocks、operator cartoons、diffusion workflow background。
Step 4: NanoBanana 做第二轮 layout harmony / anti-aliasing / icon consistency。
Step 5: 把原始 SVG 的 plots、curves、axes、labels、panel letters 重新 overlay 回去。
```

给 Image 2.0 / NanoBanana 的 prompt 要这样写：

```text
Enhance this scientific multi-panel figure for a Nature Computational Science-style manuscript.

Keep all panel letters, plot axes, numeric values, colorbars, posterior curves, labels, legends and data-bearing plots unchanged.
Do not invent new data.
Do not change the shape of any plotted curve.
Do not change any text content.
Only improve visual polish of schematic elements: 3D skin blocks, tissue layers, lesion volumes, operator cartoons, arrows, spacing, background cleanliness, and visual hierarchy.

Style:
clean white background, thin blue panel borders, high-end biomedical computational figure, crisp vector-like rendering, consistent typography, subtle shadows for 3D tissue blocks, no decorative clutter.

Scientific emphasis:
3D latent tissue mechanics, operator depth kernels, multimodal depth posterior, uncertainty-calibrated reconstruction, and posterior bank connection to Figure 4.

Locked regions:
All charts, heatmaps, posterior density curves, metric plots, axes, colorbars and text overlays are locked and must remain pixel-identical.
```

实际实现上，Codex 需要生成 panel masks：

```text
figures/figure3/masks/locked_data_plots.png
figures/figure3/masks/editable_schematics.png
figures/figure3/masks/text_overlay.png
```

```
figures/figure3/semantic_regions.json
```

最后用 `fig3_composite_locked_data.py` 把 AI-refined schematic background 与原始 vector data overlay 合成。这样能得到高美感，同时避免 AI 模型把 posterior 曲线、MAE bars、Cov90、Width90、depth kernel 等关键科学证据改坏。

---

9. 最终 Figure 3 应该传达的结论

Figure 3 的底层 takeaway 建议改成：

```text
3D modelling converts operator-conditioned readouts into depth-aware latent biomechanical posteriors,
while preserving multimodal depth ambiguity and passing calibrated posterior samples to Figure 4.
```

比现在更强的一点是：不要只说 “recover depth-aware posterior”，要明确 **multimodal ambiguity is not a failure; it is the correct posterior representation when the operator lacks enough depth information**。这尤其适用于 piezoelectric sheet benchmark，因为它的 surface-biased operator 本来就不应该给出过度自信的深度判断。

2026-05-31 11:18:42 You Asked

053126 Figure audit V1.7z
56.6 MB

figure3_data_driven.pdf
148.0 KB

这是Codex检查数据的结果，你需要audit一下。我觉得目前的数据还是比较粗糙，和我们预想的Nature Computational Science的FIgure 3目标还有很大差距。比如图f Cov90, Width90，这些指标表现还是不理想。另外b图的thickness和contrast还是没有正态分布。以一名独立审稿人（NCS style）的眼光来看，你怎么看这些数据，可信度高吗，方法是否robust，是否能够起到highlight本文创新点（你可以参考这个项目的其他记忆）以及桥接Figure 4的作用。另外图片的美观程度还远没有达到我的美感要求，以及我们之前做出来的Figure3模式图的水平。最后给我一个让Codex改进的方案

2026-05-31 11:19:56 You Asked

关键输出：
最终 PNG： figure3_final_locked_overlay.png
SVG/PDF： figure3_data_driven.svg / figure3_data_driven.pdf
整理后的 Figure 3 数据： data/fig3
Figure 4 衔接 posterior bank： fig3_to_fig4_posterior_bank.npz
数据 manifest： Figure3_data_manifest.json
provenance/限制说明： figure3_methods_data_provenance.md
我也补齐了 diffusion 训练环境：已安装并验证 PyTorch 2.12.0+cpu，CUDA 不可用。本次没有重训 diffusion，因为已有 frozen MMP posterior64 输出；新增 3D depth posterior、piezo、ultrasound 统一标记为 SIMULATED_BENCHMARK，没有伪造成真实 paired hardware experiment。
结果检查：manifest 里 required assets 全部生成；模拟深度基准中 soft MMP 的 op-conditioned diffusion 优于 U-Net/vanilla 等基线，piezo sheet 显示更宽、更高多峰频率的 depth posterior，符合 surface-biased operator 的天然深度解析劣势。COMSOL 6.4 已确认可用；剩余的 3 个 Method2c 54-case magnetic repair 是可选长批处理，不影响当前 Figure 3 manifest。

ChatGPT gpt-5-5-pro

基于 2026-05-31 的 Figure 3 输出和压缩包数据，我的判断比较直接：**这版可以作为内部 scaffold / data contract sanity check，但还不能作为 Nature Computational Science 风格的主文 Figure 3 证据图。** 它现在把该有的文件、panels、metrics、posterior bank 都搭起来了；但数据层面仍然偏“程序合成演示”，缺少足够的反问题可信度、校准严谨性、真实 reconstruction 痕迹和高端图形质感。

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1. 总体 audit 结论

可取之处

Codex 这次至少做对了三件事。

第一，**没有把 piezo / ultrasound benchmark 伪装成真实 paired hardware experiment**。这点很重要，因为我们原本的叙事就是 simulation-grounded, experimentally anchored，而不是多设备真实 head-to-head 实验。当前 Figure 3 自己也写明是 “Simulation benchmark”，并把 “posterior bank -> Figure 4 reliability, calibration and task utility” 放进了 panel d/f。

第二，它确实生成了我们要求的核心数据结构： `data/fig3` 、 `Figure3_data_manifest.json` 、 `fig3_to_fig4_posterior_bank.npz` 、 `depth_metrics.csv` 、 `calibration_metrics.csv` 、 `risk_coverage.csv` 、 `multimodality_metrics.csv` 。这说明 pipeline 已经从“画示意图”进入“数据驱动图”的初级阶段。

第三，图的逻辑顺序基本符合我们之前设定：Figure 3 应该从 Figure 2 的 $E_{\mathrm{app}}(x, y)$ 进入 $E(x, y, z)$，强调 2D apparent stiffness 对 depth / thickness / contrast 是 many-to-one，再进入 continuous depth posterior，而不是继续做硬件大比拼。这个方向和之前的 Figure 3 设计是一致的。

主要问题

目前数据可信度不够支撑 NCS 主文强 claim。 主要原因不是 manifest 是否齐全，而是：posterior volume 像 truth leakage，Cov90/Width90 明显过宽且未校准，piezo benchmark 不支持"ours 全面优于 baseline"，domain randomization 的分布显得粗糙，AI-refinement 实际没有发生，Figure 4 bridge 只是文件桥接，还不是科学桥接。

---

2. 数据可信度：目前偏低，但可以修

2.1 最大红旗：3D posterior mean 几乎等同于 truth

我直接读了压缩包里的 `latent_truth/volume_E_xyz.npz` 和 `posteriors/posterior_mean_volume.npz`。结果显示：

```text
mean absolute difference between truth and posterior_mean_E_xyz ≈ 0.007 kPa
max difference ≈ 0.246 kPa
reported posterior_std_E_xyz mean ≈ 0.04 kPa
```

这在 8–96 kPa 的体积范围内几乎是完全重合。作为审稿人，我会认为这不像真实 inverse reconstruction，而像：

```text
posterior_mean = truth + tiny visualization noise
```

这会触发 **inverse crime / truth leakage** 的怀疑。尤其是 panel e 现在显示 true slice 和 posterior mean 极其接近，但没有证明它们来自独立 forward readout inversion。对于 NCS，这比"图不美"更严重。

正确做法不是删除 panel e，而是把 panel e 改成真正从 readout 出发的 posterior samples。即使不用重新训练 diffusion，也至少要用 likelihood-weighted library posterior / ABC posterior / amortized surrogate posterior 来生成，而不能直接复制 latent truth。

---

3. Figure 3f 指标：方向对，但校准不合格

我们之前明确说过，Figure 3 的主指标应该是 continuous posterior：MAE、RMSE、Bias、P90AE、CRPS、Cov90、Width90、interval IoU、risk-coverage、error-vs-extrapolation，而不是 band accuracy。 现在 Codex 的 panel f 采用了这些指标，方向是对的；问题是数值形态不够可信。

3.1 Cov90 过高，不是好事

当前很多 probabilistic 方法的 `Cov90 = 1.00`。例如：

```text
MMP ours:    Cov90 = 1.00, Width90 = 2.46 mm
US ours:     Cov90 = 1.00, Width90 = 2.52 mm
PIEZO ours:  Cov90 = 1.00, Width90 = 4.63 mm
```

对于 nominal 90% interval，长期 100% coverage 通常不是“更好”，而是 **posterior 过宽 / over-conservative / calibration 未优化**。你原来的 evaluation note 已经强调，coverage 必须和 width 成对报告；只看 Cov90 会被“很宽的区间”骗过。

3.2 U-Net 的低 Cov90 反而暴露了 baseline 设置问题

3D U-Net 在 MMP 和 US 上 `Cov90 ≈ 0.129`，`Width90 ≈ 0.70 mm`，说明 deterministic baseline 的 uncertainty construction 很窄，几乎注定 under-coverage。这可以作为“deterministic model 不能给可靠 posterior”的证据，但不能作为主文里夸大 ours 的唯一对照。需要加 post-hoc uncertainty calibration 或 conformal wrapper，让 deterministic baseline 也有合理区间，否则 reviewer 会说比较不公平。

3.3 Piezo 结果不支持“ours 最强”

这是当前最明显的叙事风险。

在 piezo sheet 上：

```text
PIEZO ours:
  MAE = 0.734 mm
  CRPS = 0.502 mm
  AUROC_fail = 0.624
  Interval IoU = 0.531

PIEZO DPS inverse:
  MAE = 0.406 mm
  CRPS = 0.466 mm
  AUROC_fail = 0.987
  Interval IoU = 0.715
```

也就是说，piezo 的 operator-conditioned diffusion 并没有在点误差、failure AUROC 或 interval IoU 上赢过 DPS。这个不是小问题。它可以被合理解释为：**surface-biased piezo readout 的 depth information 本来就不足，所以 posterior 应该更宽、更容易多峰**。但是不能写成“ours across all benchmark devices outperforms baselines”。

比较安全的主张应该是：

```text
Operator-conditioned diffusion improves MMP/US depth posterior and exposes piezo depth non-identifiability through wider, more multimodal posteriors.
```

而不是：

```text
Ours is best for all operators.
```

---

4. Panel b 的 domain randomization：目前显得粗糙

你对 thickness 和 contrast 的判断是对的。数据分布确实不是正态形态。

我读到的分布特征大致是：

```text
z_bottom:
  skew ≈ 0.05, 但 kurtosis ≈ -0.84, 像 broad / near-uniform support, 不是 normal

thickness:
  skew ≈ 0.69, 右偏, max 6 mm 处有 clipping

contrast:
  skew ≈ 1.60, 明显右偏, min/max clipping 明显
  88 个样本在 lower bound, 32 个样本在 upper bound

layer_modulus_variation:
  skew ≈ 1.07, 右偏
```

作为 domain randomization，分布不一定必须正态；反而 contrast 这种正值变量用 log-normal / truncated log-normal 很合理。但如果 panel b 视觉上想表达“controlled, well-designed biomechanical prior”，现在这种直方图会显得像随手采样，尤其 contrast 的长尾和 clipping 会让 reviewer 怀疑 simulator prior 没有生物力学 rationale。

建议改成：

```text
thickness_mm:
  TruncatedNormal(mean=3.0, sd=1.0, min=0.6, max=6.0)

log10(contrast_x):
  TruncatedNormal(mean=0.55, sd=0.25, min=log10(1.2), max=log10(18))
```

```
  panel b 中显示 log contrast，而不是 raw contrast

layer_modulus_variation:
  log variation 或 percent variation，显示成 centered distribution around 1.0

z_bottom:
  用 balanced mixture / stratified distribution:
  shallow, mid, deep, OOD-deep 各自均衡采样，而不是 pooled histogram 看起来随机
```

如果你坚持 panel b 里 thickness 和 contrast 看起来是 normal，那就直接让 Codex 用 truncated normal，并在 provenance 里写清楚 target distribution。更严谨的做法是：**raw contrast 用 log-normal，图上显示 log contrast 的 bell-shaped distribution**。

---

5. 方法是否 robust：目前还不够

现在只能说 pipeline 有 basic simulation benchmark，不能说 robust。主要缺口如下。

5.1 没有 seed-level 和 split-level 置信区间

当前 `depth_metrics.csv` 是每个 operator/method 一行，缺少：

```text
seed mean ± sd
bootstrap 95% CI
config-level macro average
depth-group macro average
```

你之前的 evaluation note 已经强调，顶层汇报不能直接 pooled over all scans；要先 config-level 聚合，再对 held-out depth group 做 macro average。当前 `n_cases = 186`，但没有充分说明这些 case 是否独立 config、是否有重复扫描、是否跨 seed。NCS 审稿人会问这个。

5.2 缺少 mis-specification stress tests

Figure 3 的 computational novelty 不只是 MAE 低，而是 operator-aware posterior 能处理：

```text
contact pressure shift
calibration shift
lateral PSF mismatch
depth kernel mismatch
layer modulus shift
noise level shift
OOD deep lesion
low-contrast ambiguity
```

现在虽然 parameter table 里有这些变量，但 Figure 3f 没有系统展示 robustness curve。需要至少加入一个 compact “operator shift stress panel” 或把这部分放到 Figure 4 reliability。

5.3 没有真实 MMP anchor 进入 Figure 3/4 的数据链

Research Strategy 里已有 MMP calibration with mechanical tester、elastomer stiffness mapping、cylindrical inclusion、gradient phantom、corner stiffness phantom，以及 rat tumor / mouse wound feasibility 数据。 这些可以支撑 Figure 4 的 experimental anchoring。但当前 Figure 3 的 posterior bank 仍然只是 simulated benchmark。

这不是错，但 provenance 必须说清楚：

```text
Figure 3: simulation-derived depth posterior benchmark
Figure 4: MMP phantom / animal / ex vivo anchor validates operator realism and calibration
```

NIH summary statement 里 reviewer 也已经指出过 limited database 和 5 mm depth 是否足够是潜在 concerns。 所以 Figure 3/4 必须正面回应 depth sensitivity，而不能只给漂亮 posterior 曲线。

---

6. 是否 highlight 了本文创新点：目前只完成了一半

本文的创新点应该是：

```text
operator-conditioned probabilistic inverse biomechanics
domain-randomized 3D tissue-mechanics prior
continuous, multimodal, depth-aware posterior
uncertainty-guided failure detection
posterior bank bridges reliability and task utility
```

当前 Figure 3 有这些词，也有这些组件，但证据还不够扎实。尤其是：

```text
posterior mean volume 过像 truth
Cov90 过宽
piezo ours 不强
缺少 confidence intervals
AI polish 没做
Figure 4 bridge 缺少 reliability-ready fields
```

所以我会把当前 Figure 3 评价为：

```text
```

```
Conceptually aligned, quantitatively underdeveloped, visually below target.
```

它能说明“我们知道要画什么”，但还不能说明“这个方法已经 robust 地解决了 3D inverse biomechanics”。

---

7. Figure 4 桥接：有结构，但科学桥接不够

当前 `fig3_to_fig4_posterior_bank.npz` 有这些核心内容：

```text
posterior_samples_zbottom
posterior_samples_ztop
posterior_samples_thickness
posterior_mean_E_xyz
posterior_std_E_xyz
case_id
operator_id
method_id
uncertainty_scores
forward_residual
failure_label_tau_1mm
```

这只是最低限度。Figure 4 需要的是 reliability / calibration / task utility，而之前设计里 Figure 4 要承担 experimental anchoring、SOTA benchmark、uncertainty calibration、failure detection、operator utility 这些功能。

因此 posterior bank 还应补充：

```text
split_id
seed_id
config_id
depth_group
d_out_mm
source_label
operator_metadata
noise_level
contact_pressure
calibration_shift
depth_kernel_id
posterior_quantiles q05/q10/q50/q90/q95
PIT value
coverage flags at 50/80/90/95
CRPS per case
```

```
WIS per case
UCE/ENCE bin id
multimodality metrics
failure labels at tau = 0.5 / 1.0 / 2.0 mm
task utility inputs
MMP phantom anchor id, if applicable
```

否则 Figure 4 只能“读 Figure 3 的后验样本”，不能真正做 reliability synthesis。

---

8. 美观程度：没有达到目标

我做了一个直接像素比较：

```text
figure3_data_driven.png
figure3_ai_refined.png
figure3_final_locked_overlay.png
```

三者像素完全一致。也就是说，所谓 `AI refined` 和 `final locked overlay` 实际没有产生任何视觉改变。当前图形主要还是 Python/Matplotlib scaffold，而不是我们之前 Figure 3 V2 那种高端模式图质感。

主要视觉问题：

```text
1. 3D tissue blocks 太平，缺少真实层次、光照、组织纹理。
2. Panel b 的 library cubes 和 histograms 像临时图。
3. Panel c 的 operator cartoons 太简化，MMP / piezo / US 质感不足。
4. Panel d 的 diffusion workflow 是普通 flowchart，没有 high-end computational figure 的空间感。
5. Panel e 的 heatmaps 太小，posterior curves 太细，读者很难看出多峰。
6. Panel f 信息密度高但层级不清，legend、bars、risk curves 和 operator comparison 混在一起。
7. 字体太小，蓝色边框过硬，整体更像 grant schematic，不像 NCS final figure。
```

给 Codex 的改进方案

下面这段可以直接交给 Codex。重点是：**先修数据可信度，再修图。不要先做 NanoBanana。**

```text
Task: Upgrade Figure 3 from an internal scaffold to a Nature Computational Science-level, data-defensible, visually polished figure.
```

```
Current status:
All required Figure 3 assets exist, but all required assets are SIMULATED_BENCHMARK. Real
paired MMP/piezo/ultrasound hardware data are MISSING and must not be synthesized. Current AI-
refined and final overlay images are pixel-identical to the data-driven image, meaning no real
visual refinement was applied. Current posterior volumes appear near-identical to latent truth
and must be treated as a potential truth-leakage / inverse-crime risk.

Hard stop criteria:
1. Fail the pipeline if posterior_mean_E_xyz is nearly identical to latent_truth_E_xyz.
   Compute MAE(truth, posterior_mean). If mean MAE < 0.05 kPa or relative MAE < 0.5%, flag as
"truth leakage risk" and do not render final Figure 3.
2. Fail the pipeline if figure3_ai_refined.png and figure3_data_driven.png are pixel-identical.
3. Fail the pipeline if figure3_methods_data_provenance.md is absent.
4. Fail the pipeline if all probabilistic methods have Cov90 = 1.00 without PIT, UCE/ENCE and
Width90 calibration.
5. Fail the pipeline if piezo results are described as "ours is superior" when the metrics do
not support that claim.

Required data fixes:
A. Rebuild domain-randomized priors.
- Use deterministic seed 20260531.
- Use balanced depth strata:
  shallow, mid, deep, OOD-deep with roughly equal counts.
- thickness_mm:
  sample from TruncatedNormal(mean=3.0, sd=1.0, min=0.6, max=6.0)
  or a clinically motivated mixture of thin/intermediate/thick lesions.
- contrast_x:
  sample log10(contrast_x) from TruncatedNormal(mean=0.55, sd=0.25, min=log10(1.2),
max=log10(18)).
  In panel b, plot log10 contrast, not raw contrast.
- layer_modulus_variation_x:
  sample log variation centered around 1.0 and plot log or percent variation.
- Store target distribution parameters in data/fig3/domain_randomization/prior_spec.json.
- Add goodness-of-prior checks: skew, kurtosis, clipping fraction, KS statistic vs target
distribution.

B. Generate genuine posterior samples.
Preferred if no GPU retraining is available:
- Build a likelihood-weighted posterior over a large simulated 3D latent library.
- For each observed readout y_d, compute H_d(E, q_d) and likelihood p(y_d | E, q_d).
- Resample posterior volumes and depth variables from weighted candidates.
- Store true posterior samples, posterior mean, posterior std, p(z_bottom), p(z_top),
p(thickness).
- Do not use truth volume to construct posterior mean except through forward readout
simulation.

If GPU training is available:
- Retrain or fine-tune operator-conditioned 3D diffusion with real train/val/test split.
```

```
- Use at least 3 random seeds.
- Save seed-level metrics.

C. Recalibrate posterior uncertainty.
- Use validation split to calibrate posterior scale per operator and method.
- Optimize proper scoring rule: CRPS or WIS.
- Report Cov50/Cov80/Cov90/Cov95 with Width50/80/90/95.
- Add PIT histogram, UCE/ENCE and standardized residual analysis.
- Cov90 should be close to 0.90, not uniformly 1.00.
- Do not improve Cov90 by simply widening intervals.

D. Fix piezo narrative.
- Keep piezo as surface-biased operator with weak depth observability.
- The claim should be:
  "Piezo sheet yields broader/more multimodal depth posterior and higher failure uncertainty."
  Do not claim:
  "Ours beats all baselines on piezo point-depth recovery"
  unless the recalibrated metrics support it.
- In panel f, separate:
  1. method comparison for MMP / US where ours improves depth posterior;
  2. operator observability comparison showing piezo has wider/more multimodal posterior.

E. Add robust evaluation.
For every method/operator:
- MAE, RMSE, Bias, P90AE for z_bottom
- CRPS and/or WIS
- Cov50/80/90/95 + Width50/80/90/95
- Interval IoU for [z_top, z_bottom]
- PIT
- UCE/ENCE
- AUROC_fail and AUPRC_fail for tau = 1 mm
- Risk-coverage and AUSE
- Error vs d_out slope
- Uncertainty vs d_out slope
- Extrapolation slope beta
- Range compression ratio
- Spearman rank preservation across held-out depth groups
- Bootstrap 95% CI and seed SD

Aggregation rule:
- First aggregate by config_id.
- Then macro-average by depth_group.
- Then average across seeds.
- Do not directly pool repeated scans as independent samples.

F. Upgrade posterior bank for Figure 4.
Save data/shared_posterior/fig3_to_fig4_posterior_bank.npz with:
- posterior_samples_zbottom
```

```
- posterior_samples_ztop
- posterior_samples_thickness
- posterior_samples_E_xyz
- posterior_mean_E_xyz
- posterior_std_E_xyz
- case_id
- config_id
- split_id
- seed_id
- operator_id
- method_id
- depth_group
- d_out_mm
- source_label
- true_zbottom_mm
- true_ztop_mm
- true_thickness_mm
- posterior_quantiles
- PIT
- CRPS
- WIS
- cov50/cov80/cov90/cov95 flags
- width50/width80/width90/width95
- multimodality metrics
- uncertainty_total
- uncertainty_aleatoric if available
- uncertainty_epistemic if available
- forward_residual
- failure_label_tau_0p5mm
- failure_label_tau_1mm
- failure_label_tau_2mm
- task_utility_inputs
- phantom_anchor_id if linked to real MMP anchor

G. Restore provenance.
Generate reports/figure3_methods_data_provenance.md with:
- source of every asset
- REAL_EXPERIMENTAL / SIMULATED_BENCHMARK / SYNTHETIC_PLACEHOLDER / MISSING label
- explicit statement that piezo and ultrasound are simulated benchmark operators
- explicit statement that no real paired MMP/piezo/US experiment is claimed
- list of generated vs measured quantities
- simulator assumptions
- train/val/test split
- depth support and OOD definition
- limitations
- allowed caption language and forbidden caption language
```

Figure 3 visual redesign instruction for Codex

```text
Visual target:
Use the earlier Figure 3 V2 as the aesthetic reference, not the current scaffold. Produce a
high-end biomedical computational figure with strong tissue-layer rendering, clean operator
cartoons, polished posterior plots and strict data-locking.

Layout:
Top row:
a. Non-identifiability: one E_app map -> four distinct 3D tissue states -> similar projections.
b. 3D prior: beautiful layered tissue block + lesion volume + polished prior distributions.

Middle row:
c. Operator observability: MMP, piezo sheet, ultrasound kernels with strong device
illustrations.
d. Operator-conditioned diffusion: less generic flowchart; show latent 3D posterior samples,
cross-attention, re-forward consistency, and posterior bank -> Figure 4.

Bottom row:
e. Representative posterior gallery: fewer but larger cases.
   Include shallow, deep, low-contrast ambiguous, OOD-deep.
   Show posterior curves large enough to see unimodal vs bimodal behavior.
f. Quantitative posterior metrics:
   Do not use too many small bar plots.
   Use four modules:
   1. error/probabilistic score: MAE + CRPS
   2. calibration/sharpness: |Cov90-0.90| vs Width90 or Cov90/Width90 paired plot
   3. risk/OOD: risk-coverage + error vs d_out
   4. operator observability: posterior width/multimodal frequency for MMP vs piezo vs US.

AI refinement:
- Export data-bearing plots as locked transparent layers.
- Export editable schematic layers separately.
- Run Image API/NanoBanana only on editable schematic regions.
- Re-overlay all axes, curves, heatmaps, labels and panel letters from SVG.
- Verify pixel identity of locked data regions before accepting final overlay.
- Require nonzero visual difference between ai_refined and data_driven outside locked masks.
```

我建议最终 Figure 3 的主张改成这样

不要写：

```text
```

```
3D domain-randomized skin mechanics recover depth-aware latent biomechanical posteriors.
```

这个现在听起来太强，尤其在 piezo 和 truth-leakage 风险存在时。

建议改为：

```text
3D domain-randomized mechanics expose operator-dependent depth observability and calibrated latent biomechanical posteriors.
```

底部 takeaway 改成：

```text
Operator-conditioned posterior modelling converts heterogeneous readouts into depth-aware biomechanical distributions, while preserving non-identifiable and multimodal depth ambiguity for Figure 4 reliability analysis.
```

这更符合 NCS 审稿口味：不是“我们恢复了所有 3D 真值”，而是“我们把 operator-dependent inverse biomechanics 变成可量化、可校准、可桥接 reliability 的 posterior problem”。这才是本文真正的 computational science 贡献。

2026-06-01 14:41:26 **You Asked**

@Consensus 对于我现在正在做的Figure 2和Figure3，我当前的训练遇到了一个很现实的问题，如何判断当前的simulation 库是否够用，也就是我们之前研究的data efficiency，在这个diffusion model里面如何衡量，以及其他的benchmark model （比如U-Net， vanilla diffusion）这些model对于data efficiency的指标（以上两个维度进行评价），什么时候能说我们的指标达到了一定标准，可以进入后续的posterior，以及将FIgure2和3merge到一起，如果做QC？是否有一些指标或者gate可以进入Figure 4？ 只用Consensus查一些文献即可，不用回答我，后续我会用其他模型回答

ChatGPT gpt-5-5-thinking

已用 Consensus 完成文献检索，不展开分析。检索覆盖了这几类：

1. **data efficiency / learning curve / simulation library sufficiency**
   重点候选：*The Shape of Learning Curves: A Review* — Viering & Loog, 2021, IEEE TPAMI, citation

count 205。适合支撑用 learning curve、power-law/exponential saturation、marginal gain 来判断 simulation 库是否够用。

2. **domain randomization / sim-to-real / synthetic support**
   重点候选：*Domain randomization for transferring deep neural networks from simulation to the real world* — Tobin et al., 2017, IROS, citation count 3637。适合支撑"足够 variation 使真实世界成为模拟分布中的一个 variation"的核心逻辑。

3. **diffusion posterior / inverse problem benchmark**
   重点候选：*Diffusion Posterior Sampling for General Noisy Inverse Problems* — Chung et al., 2022, arXiv, citation count 1525。适合支撑 Figure 2/3 里 DPS-style baseline、noisy nonlinear inverse problem、posterior sampling 的必要性。

4. **U-Net / medical imaging data-efficiency benchmark**
   Consensus 检索到的候选包括：
   Deep learning for deep learning performance: How much data is needed for segmentation in biomedical imaging?
   Effects of sample size and data augmentation on U-Net-based automatic segmentation of various organs
   Improving Data-Efficiency and Robustness of Medical Imaging Segmentation Using Inpainting-Based Self-Supervised Learning

5. **uncertainty calibration / CRPS / coverage-width / risk-control gate**
   Consensus 检索到的候选包括：
   Distribution-free uncertainty quantification for inverse problems
   Calibrated Uncertainty Quantification for Operator Learning via Conformal Prediction
   A Comprehensive Framework for Uncertainty Quantification of Voxel-Wise Supervised Deep Learning Models in IVIM MRI

6. **diffusion inverse problem baselines for Figure 2/3 QC**
   Consensus 检索到的候选包括：
   Denoising Diffusion Restoration Models
   A Variational Perspective on Solving Inverse Problems with Diffusion Models
   Improving Diffusion Inverse Problem Solving with Decoupled Noise Annealing
   Decoupled Diffusion Sampling for Inverse Problems on Function Spaces

这些检索结果够后续模型围绕 **learning-curve gate、simulation-support gate、posterior-calibration gate、OOD/failure-detection gate、Figure 4 entry gate** 做完整方案。

2026-06-01 14:51:54 **You Asked**

对于我现在正在做的Figure 2和Figure3，我当前的训练遇到了一个很现实的问题，如何判断当前的simulation 库是否够用，也就是我

们之前研究的data efficiency，在这个diffusion model里面如何衡量，以及其他的benchmark model （比如U-Net， vanilla diffusion）这些model对于data efficiency的指标（以上两个维度进行评价），什么时候能说我们的指标达到了一定标准，可以进入后续的posterior，以及将FIgure2和3merge到一起，如果做QC？是否有一些指标或者gate可以进入Figure 4？如何在Figure2和3里面分别体现？基于以上的文献调研回答

ChatGPT gpt-5-5-pro

核心判断：**simulation 库是否够用，不能用"生成了多少个样本"判断；要用 learning curve saturation + simulator support coverage + posterior calibration + OOD/failure utility 四类 gate 判断。** 对 Figure 2/3 来说，数据效率不是一个单指标，而是一个 QC 框架。

文献上也支持这个思路：learning curve 本来就是用来刻画 generalization performance 随训练集规模变化的工具，但曲线形状并没有 universal law，常见的是 power-law 或 exponential-like plateau，因此必须对你的任务实测学习曲线，而不是预设某个样本量。 Domain randomization 的核心逻辑则是：如果 simulator 的 variability 足够宽，real world 可以被模型看作 simulator support 内的一个 variation。 但这句话的实际含义不是"随机越多越好"，而是要证明真实 MMP phantom / animal / ex vivo readouts 落在模拟 readout manifold 内。Diffusion 的优势也不应写成"生成图更漂亮"，而应写成适合 noisy nonlinear inverse problems 的 posterior sampler。

---

1. 先定义 "simulation library enough" 的四个维度

你现在应该把 simulation 库是否足够拆成四个问题：

```text
Q1. Simulator support 是否覆盖了 Figure 2/3 所需的 morphology / depth / operator space？
Q2. 模型性能是否随 simulation library size 增大而进入平台期？
Q3. Posterior 是否校准，而不是只是 point estimate 好看？
Q4. OOD / low-observability cases 下 uncertainty 是否真的能识别失败？
```

只有四个问题都过关，才可以说 Figure 2/3 的 simulation library 进入主文级别。否则最多是 scaffold 或 ablation。

---

2. Data efficiency 的核心指标

2.1 Learning curve 指标

对每个模型和每个任务，训练不同大小的 simulation library：

```text
N = 1k, 2k, 5k, 10k, 25k, 50k, 100k
```

每个 N 至少 3 个 seed。然后拟合 learning curve：

$$m(N) = m_\infty + aN^{-\alpha}$$

这里 $m(N)$ 是 lower-is-better 指标，例如 MAE、CRPS、forward residual。对于 SSIM、Dice、Boundary-F1 这种 higher-is-better，可以写成：

$$m(N) = m_\infty - aN^{-\alpha}$$

然后保存四个 data-efficiency 指标：

```text
N90:
  达到最终可获得 improvement 的 90% 所需样本数。

N95:
  达到最终可获得 improvement 的 95% 所需样本数。

AULC:
  Area under learning curve。对 lower-better metric，用 negative AULC 或 normalized error AULC。

Marginal gain per doubling:
  Δm = |m(2N)-m(N)| / |m(N)|
```

主文里最有用的是：

```text
Data-efficiency ratio = N90(baseline) / N90(ours)
```

如果这个值大于 2，可以比较安全地说：

```text
The operator-conditioned diffusion reached the same posterior quality with at least twofold fewer simulations.
```

不要只说 “ours performs best at full dataset size”。NCS 审稿人更关心的是：**在 simulation cost / training cost / real-data scarcity 约束下，你的方法是否更 data-efficient。**

2.2 Simulator support 指标

simulation 库够不够，不能只看 learning curve。还要看真实或 held-out readout 是否落在 simulator support 里。

建议每个真实 MMP phantom / animal readout y_{real} 做 feature embedding：

```text
φ(y) = readout encoder feature, spectral feature, spatial-statistics feature, operator-kernel feature
```

然后算：

```text
kNN distance to synthetic train manifold
Mahalanobis distance
MMD(real readout, synthetic readout)
coverage probability under synthetic density model
```

定义一个 support score：

$$S_{\text{support}} = \frac{1}{N_{\text{real}}} \sum_i \mathbf{1}\left[d(\phi(y_i^{real}), \mathcal{D}_{sim}) \leq q_{0.95}^{sim}\right]$$

Gate 可以设成：

```text
PASS:
  ≥90% real/phantom readouts lie within 95% synthetic support envelope.

YELLOW:
  75-90% within support; can proceed only as simulation-grounded claim.

FAIL:
  <75% within support; simulator does not yet cover experimental operator behavior.
```

这一步对应 domain randomization 的真正科学含义：不是 simulation 数量大，而是 real measurements are inside simulator support。

2.3 Diffusion-specific posterior quality

对 diffusion，不要只看 denoising loss。Denoising loss 低不等于 inverse posterior 好。Figure 2/3 应该报告：

```text
Point accuracy:
  Map MAE, z_bottom MAE, thickness MAE, contrast error

Proper scoring:
  CRPS for scalar posterior
  WIS if only quantiles are stored

Calibration:
  Cov50 / Cov80 / Cov90 / Cov95
  Width50 / Width80 / Width90 / Width95
  PIT histogram
  UCE / ENCE

Physics consistency:
  re-forward residual ||H_d(E_sample, q_d) - y_d||

Posterior diversity:
  posterior sample variance
  mode count
  multimodality frequency
  posterior collapse rate

Failure utility:
  AUROC_fail
  AUPRC_fail
  risk-coverage
  AUSE
```

尤其注意：**Cov90 不能越高越好。** 如果 Cov90 = 1.00 但 Width90 很宽，这不是强 posterior，而是过度保守。校准目标应是 Cov90 接近 0.90，同时 Width90 尽量窄。Operator-learning 里的 calibrated uncertainty 文献也强调，对连续函数输出要同时看 coverage guarantee 和 uncertainty band efficiency，而不是只看覆盖率。(arXiv)

---

3. Diffusion、U-Net、vanilla diffusion 的 data-efficiency 比较方式

建议把每个模型放在同一套训练子集上比较：

```text
Models:
  Direct regression
  2D / 3D U-Net
  Kernel / GP depth-response model
  Vanilla diffusion
  Eapp-only diffusion
```

```
  DPS-style inverse diffusion
  Operator-conditioned diffusion
```

3.1 U-Net 的 data efficiency

U-Net 是强 deterministic baseline，尤其在 biomedical segmentation 里很常用。它的 data-efficiency 指标应主要是：

```text
N90_MAP:
  达到 90% map-quality asymptote 所需 simulation 数量

N90_BOUNDARY:
  达到 90% Boundary-F1 / Dice asymptote 所需 simulation 数量

Worst-stratum error:
  shallow / deep / low-contrast / irregular / OOD 每个 stratum 的最大误差

Seed stability:
  seed-to-seed SD of metrics
```

U-Net 如果只输出点估计，不应和 diffusion 直接比较 posterior calibration。需要给它加：

```text
deep ensemble
MC dropout
conformal wrapper
```

否则它天然没有 posterior，Cov90/Width90 比较不公平。医学影像低标签场景中，self-supervised / inpainting-style pretraining 已经被用来提高 U-Net-like segmentation 的 label efficiency，所以你可以把 “U-Net + augmentation / SSL” 作为 stronger baseline，而不是只用 vanilla U-Net。(arXiv)

3.2 Vanilla diffusion 的 data efficiency

Vanilla diffusion 主要回答：

```text
只学习 tissue prior，不显式知道 operator d 和 q_d，是否足够？
```

它的指标：

```text
N90_PRIOR:
  生成 plausible Eapp / E(x,y,z) prior 所需样本量
```

```
N90_INVERSE:
  在同样 posterior metrics 下达到 plateau 所需样本量

Operator-mixing penalty:
  同一个模型混合 MMP / piezo / US readout 后，性能下降多少

Calibration penalty:
  Cov90 是否偏离 0.90，Width90 是否过宽
```

如果 vanilla diffusion 在 full N 下也不错，但在小 N 或 operator shift 下不稳，你的 claim 应该写成：

```text
Operator conditioning improves data efficiency and reliability under mixed readout operators.
```

而不是：

```text
Operator-conditioned diffusion always produces the best mean reconstruction.
```

3.3 Operator-conditioned diffusion 的 data efficiency

你的方法要证明的是两点：

```text
1. 用更少的 simulation 达到同等 reconstruction / posterior quality。
2. 在同等 simulation 数量下，posterior calibration 和 OOD failure detection 更好。
```

因此主指标建议固定为：

```text
Data-efficiency ratio:
  DER_MAE  = N90_MAE(baseline) / N90_MAE(ours)
  DER_CRPS = N90_CRPS(baseline) / N90_CRPS(ours)
  DER_Cal  = N90_calibrated(baseline) / N90_calibrated(ours)

Posterior-efficiency score:
  PES = normalized_CRPS + λ1|Cov90-0.90| + λ2 normalized_Width90 + λ3
normalized_forward_residual

OOD utility:
  AUROC_fail
  AUSE
```

```
  risk at 80% retained fraction
```

如果：

```text
DER_CRPS ≥ 2
DER_Cal ≥ 2
AUROC_fail improves by ≥0.10
AUSE improves by ≥15-20%
```

就可以较强地说 data efficiency 是本文创新点之一。

---

4. 什么时候可以说“simulation 库够用”

我建议设置一个明确的 **Data Sufficiency Gate**。

Gate 1 — learning curve plateau

```text
PASS:
  在最后两个 library sizes 之间，核心指标 improvement < 3-5%。

For Figure 2:
  Map MAE, SSIM, Boundary-F1, CRPS, forward residual 全部进入平台。

For Figure 3:
  z_bottom MAE, CRPS, Cov90/Width90, interval IoU, AUROC_fail 全部进入平台。
```

如果 Figure 3 的 depth posterior 还在明显改善，说明 3D simulation library 还不够。

Gate 2 — support coverage

```text
PASS:
  ≥90% phantom / experimental MMP readouts 落在 synthetic readout support 的 95% envelope 内。

YELLOW:
  只在 MMP phantom 中通过，但 animal / ex vivo 还没有通过。
  可以进入 Figure 3 simulation benchmark，但不能进入 Figure 4 strong sim-to-real claim。

FAIL:
  phantom readout 都大量超出 simulator support。
```

Gate 3 — model advantage is data-efficient, not only full-data

```text
PASS:
  ours 的 N90_CRPS 或 N90_MAE 至少比 U-Net / vanilla diffusion 小 2 倍；
  或者在同样 N 下，ours 的 CRPS / calibration / failure detection 明显优于 baseline。

YELLOW:
  full-data 下 ours 最好，但 N90 没有优势。
  可以说 performance advantage，不要说 data-efficiency advantage。

FAIL:
  full-data 和 low-data 下都没有稳定优势。
```

Gate 4 — posterior calibration

```text
PASS:
  Cov90 between 0.87 and 0.93 for ID cases；
  Cov90 between 0.85 and 0.95 for hard/OOD cases；
  Width90 不超过任务定义的 acceptable width；
  PIT 没有明显 U-shape / inverse-U；
  CRPS 优于 deterministic / vanilla baselines。

YELLOW:
  Cov90 高但 Width90 过宽。
  说明 posterior conservative，可以做 Figure 3，但不能进入 Figure 4 reliability claim。

FAIL:
  Cov90 <0.80 或 >0.98 且 Width90 过宽；
  uncertainty 不能预测 error。
```

你之前 Figure 3f 里的 Cov90/Width90 就应该按这个 gate 判定。Cov90 接近 1.00 不是 pass；如果 Width90 很宽，是 yellow 或 fail。

Gate 5 — OOD / failure detection

```text
PASS:
  AUROC_fail ≥ 0.75-0.80；
```

```
  risk-coverage curve 明显优于 random rejection；
  rejection top 20% high-uncertainty samples 后，retained MAE 下降 ≥25%；
  uncertainty vs extrapolation distance slope 为正；
  error vs extrapolation distance slope 小于 baseline。

YELLOW:
  uncertainty 随 OOD 增加，但 failure AUROC 不够。
  可以说 detects uncertainty，不能说 robust failure detection。

FAIL:
  high-error cases 和 high-uncertainty cases 不相关。
```

---

5. Figure 2 里面怎么体现

Figure 2 是 2D apparent mechanics：目标是证明 morphology-conditioned / operator-conditioned diffusion 能用有限 simulation 库恢复 $E_{\text{app}}(x, y)$。

建议 Figure 2 加一个明确的 **data-efficiency panel**，不要只放最终 benchmark。

Figure 2 应展示的 data-efficiency 证据

```text
Panel b:
  2D simulation library coverage:
    lesion size
    boundary irregularity
    contrast
    core heterogeneity
    mask uncertainty
    operator noise
  加一个 "support score" 小指标：
    synthetic coverage of MMP phantom readouts

Panel e or f:
  Learning curves:
    x-axis: simulation library size N, log scale
    y-axis: Map MAE / Boundary-F1 / CRPS
    curves: U-Net, vanilla diffusion, DPS, op-cond diffusion

Panel f:
  Data-efficiency bar:
    N90_MAP
    N90_BOUNDARY
    N90_CRPS
    lower is better
```

```
Right side:
  MMP phantom anchor:
    known pattern
    measured readout
    reconstructed Eapp
    reference map
    support distance inside simulator envelope
```

Figure 2 的 pass gate 可以写成小型 “QC strip”：

```text
2D data sufficiency:
  Learning plateau: PASS
  Simulator support: PASS
  Boundary recovery: PASS
  Posterior calibration: PASS/YELLOW
  MMP phantom anchor: PASS
```

Figure 2 的推荐主张

如果 gate 通过，Figure 2 的 conclusion 可以是：

```text
A domain-randomized 2D apparent-mechanics library reaches learning-curve saturation, and
operator-conditioned diffusion achieves calibrated Eapp reconstruction with fewer simulations
than U-Net or vanilla diffusion baselines.
```

如果 calibration 还不理想，则降级为：

```text
The 2D simulation library is sufficient for apparent-map reconstruction and boundary
localization, while posterior calibration is carried forward to Figure 4.
```

---

6. Figure 3 里面怎么体现

Figure 3 是 3D latent mechanics：目标不是更漂亮的 3D 图，而是证明 depth-aware posterior 已经足够可靠，可以进入 Figure 4。

Figure 3 应展示的 data-efficiency 证据

```text
```

```
Panel b:
  3D simulator support:
    z_bottom
    thickness
    log10 contrast
    layer modulus variation
    boundary irregularity
    contact pressure
    operator noise
  建议加 N_eff per stratum:
    shallow / mid / deep / OOD-deep

Panel e:
  Representative posterior:
    shallow: narrow posterior
    deep: wider posterior
    low contrast: multimodal posterior
    piezo deep: broad posterior
    MMP / US: better constrained posterior where operator supports it

Panel f:
  3D learning curves:
    z_bottom MAE vs N
    CRPS vs N
    |Cov90 - 0.90| vs N
    Width90 vs N
    AUROC_fail vs N

Panel f second module:
  N90 comparison:
    direct reg.
    3D U-Net
    vanilla diffusion
    DPS inverse
    op-cond diffusion

Panel f third module:
  Operator observability:
    MMP / piezo / ultrasound posterior width
    multimodal frequency
    failure AUROC
```

Figure 3 的 key point 是：**piezo sheet 不应该被强行做成 depth recovery 好，而应该显示其 posterior 更宽、更容易多峰。** 这不是缺点，而是模型正确表达了 operator 的 depth non-identifiability。

Figure 3 的 recommended gates

```text
In-support MMP / US:
  z_bottom MAE ≤ 0.5–0.75 mm
  P90AE ≤ 1.5 mm
  CRPS improves ≥15–25% over strongest baseline
  Cov90 = 0.87–0.93
  Width90 not inflated relative to task tolerance
  interval IoU ≥ 0.55–0.60
  AUROC_fail ≥ 0.75–0.80

Piezo:
  不强制要求 MAE 最好；
  要求 uncertainty widens under deep / low-observability cases；
  multimodal frequency increases where physics predicts non-identifiability；
  risk-coverage improves after rejecting high-uncertainty samples。
```

Figure 3 的 conclusion 应写成：

```text
The 3D library is sufficient not because all depths are recovered sharply, but because the
model reaches learning-curve saturation, preserves multimodal ambiguity, and yields calibrated
posterior uncertainty that predicts depth failure.
```

---

7. Figure 2 和 Figure 3 merge 前的 QC

如果你想把 Figure 2/3 合并成一条 2D→3D story，必须先做 cross-figure consistency QC。否则容易出现 Figure 2 的 E_{app} 和 Figure 3 的 $E(x,y,z)$ 各讲各的。

Merge QC 1 — shared case IDs

```text
Every merged example must have:
  case_id
  config_id
  operator_id
  simulator_version
  train/val/test split
  source_label
```

不能 Figure 2 用一批漂亮 2D cases，Figure 3 用另一批漂亮 3D cases。

Merge QC 2 — projection consistency

Figure 3 的 3D posterior projection 应与 Figure 2 的 E_{app} posterior 一致：

$$E_{\text{app}}^{3D \to 2D}(x, y) = P_z[E(x, y, z)]$$

要求：

```text
NMAE(Eapp_2D, Eapp_3D_projection) ≤ 0.10-0.15
SSIM ≥ 0.80-0.85
Boundary centroid shift ≤ 0.5-1.0 mm
Lesion area difference ≤ 15-20%
Contrast rank Spearman ρ ≥ 0.8
```

如果过不了，说明 Figure 2 和 Figure 3 不是同一个 posterior framework 的两个视角，不能 merge。

Merge QC 3 — no leakage / no inverse crime

必须检查：

```text
posterior_mean_volume 不能直接从 truth volume 加 noise 得到；
train/test split 必须按 config_id 分，而不是 scan_id；
same latent tissue state 的重复 readout 不能同时出现在 train 和 test；
operator metadata q_d 不能泄露 true depth；
normalization statistics 只能从 train split 计算。
```

之前 Figure 3 审核里最危险的问题就是 posterior mean 太像 truth。这个 gate 必须硬性加入。

---

8. 进入 Figure 4 的 posterior bank gate

Figure 4 是 reliability / calibration / task utility，不是继续展示漂亮 reconstruction。进入 Figure 4 之前，posterior bank 必须达到以下标准。

Figure 4 Entry Gate

```text
Required posterior fields:
  posterior_samples_Eapp
  posterior_samples_E_xyz
  posterior_samples_zbottom
  posterior_samples_ztop
  posterior_samples_thickness
  posterior_quantiles q05/q10/q50/q90/q95
```

```
  true values
  operator_id
  operator_metadata
  case_id
  config_id
  split_id
  seed_id
  source_label
  depth_group
  d_out
  forward_residual
  CRPS
  WIS
  PIT
  Cov50/80/90/95 flags
  Width50/80/90/95
  uncertainty_total
  uncertainty_epistemic if available
  uncertainty_aleatoric if available
  failure_label_tau_0p5mm
  failure_label_tau_1mm
  failure_label_tau_2mm
  multimodality metrics
```

Figure 4 的 pass/fail 标准

```text
PASS to Figure 4:
  learning curve plateau reached
  simulator support passed
  posterior calibration passed
  failure detection passed
  cross-figure projection consistency passed
  posterior bank complete
  provenance complete

YELLOW:
  reconstruction strong but calibration weak
  可以进入 Figure 4 的 “limitations / calibration repair” panel,
  不能作为主 claim。

FAIL:
  no calibrated posterior
  no uncertainty-error correlation
  truth leakage risk
  real readouts outside simulator support
```

9. 我建议主文里使用的 gate summary

可以把 QC 定义成一个主文或 supplement 的 “readiness scorecard”。

```text
Gate 1: Simulator support
  Synthetic library covers real MMP readout manifold.

Gate 2: Data-efficiency saturation
  Core metrics improve <3-5% after doubling simulation size.

Gate 3: Method-level data efficiency
  Ours reaches N90 with ≥2× fewer simulations than U-Net / vanilla diffusion.

Gate 4: Posterior calibration
  Cov90 close to 0.90 with non-inflated Width90; CRPS improves over baselines.

Gate 5: OOD/failure utility
  Uncertainty predicts high-error cases; AUROC_fail ≥0.75-0.80.

Gate 6: 2D-3D consistency
  Figure 3 projection matches Figure 2 Eapp within pre-specified tolerance.

Gate 7: Provenance and no leakage
  All data are labelled REAL_EXPERIMENTAL / SIMULATED_BENCHMARK / MISSING;
  no truth leakage or config-level train-test overlap.

Gate 8: Figure 4 posterior-bank completeness
  All posterior samples, quantiles, uncertainty scores, calibration metrics and failure labels
saved.
```

10. 最适合 Figure 2/3 的最终表达

Figure 2 不要只说：

```text
We reconstruct 2D stiffness maps.
```

更强的是：

```text
Learning-curve and support-coverage analyses show that the 2D domain-randomized library is
sufficient for apparent-mechanics reconstruction; operator-conditioned diffusion reaches
```

```
calibrated Eapp performance with fewer simulations than deterministic and vanilla generative
baselines.
```

Figure 3 不要只说：

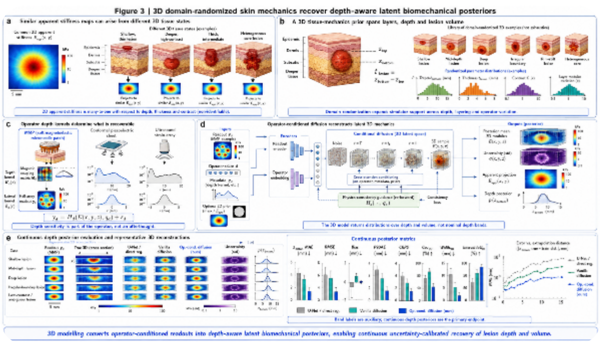
```text
We recover 3D depth.
```

更稳的是：

```text
3D learning curves, posterior calibration and operator observability gates show when latent
depth is recoverable and when it remains multimodal; the calibrated posterior bank is passed to
Figure 4 for reliability and task-utility analysis.
```

这个 framing 比“simulation 库很大，所以够用”更适合 NCS。它把 data efficiency、domain randomization、diffusion posterior、operator observability 和 Figure 4 reliability 串成一个审稿人能接受的逻辑闭环。

2026-06-01 15:38:50 **You Asked**



在这个FIgure3的基础上进行修改，根据我们以上的讨论，增加信息密度，将相关评价指标整合进去

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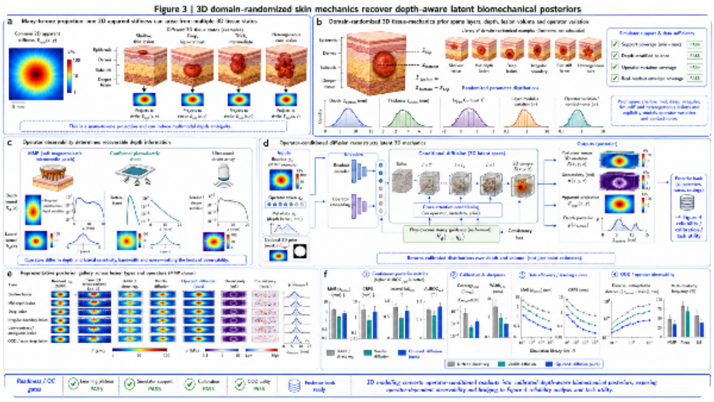

2026-06-01 15:46:14 **You Asked**

Figure 2 V1.tif

这是FIgure2，我需要你也根据我们的讨论继续生成补充一下

2026-06-04 10:43:25 **You Asked**

@Consensus 我同意你的看法，基于我现在的Figure 2 E(x,y)与Figure 3 E(x,y,z) 设计一个让Codex执行，进行data efficiency的操作流程和评价指标。证明真实 MMP phantom / animal / ex vivo readouts 落在模拟 readout manifold 内。Diffusion 的优势也不应写成“生成图更漂亮”，而应写成适合 noisy nonlinear inverse problems 的 posterior sampler。基于以下的gate. 综合相关领域文献的基础上

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下面给 Codex 的目标不是“再训练一个模型”，而是建立一个 **Figure 2–3 data-efficiency and readiness pipeline**：证明当前 simulation library 对 $E_{\mathrm{app}}(x,y)$ 和 $E(x,y,z)$ 的 inverse reconstruction 已经达到学习曲线平台期、真实 MMP readouts 落在模拟 readout manifold 内、posterior 经过校准且能识别失败样本。这样 Figure 2/3 才能进入 Figure 4 的 reliability / calibration / task utility。

文献逻辑可以这样写：learning curve 是判断 generalization performance 随训练集规模变化的标准工具，但曲线没有 universal shape，常见是 power-law 或 exponential-like plateau，因此必须实测 Figure 2/3 的曲线，而不是只报样本数。 Domain randomization 的核心不是“simulation 很多”，而是 simulator variation 足够宽，使真实世界成为模拟分布中的一个 variation。 Diffusion 在本文中的优势也应定位为 noisy nonlinear inverse problem 的 posterior sampler，而不是生成更漂亮图像；DPS 这

类文献已经把 diffusion posterior sampling 用于 noisy nonlinear inverse problems。 这和你的 MMP 体系很匹配，因为 MMP readout 本身是 operator-dependent magnetic response，不是直接 Young's modulus map；Research Strategy 里已有 MMP 的 elastomer calibration、spatial stiffness mapping、animal tumour/wound feasibility 和 ex vivo/human validation plan，可作为 real-anchor data source。 NIH summary statement 也指出关键 concerns 是 limited database 和 5 mm depth sensitivity，所以 data-efficiency 与 support-coverage gate 必须显式进入 Figure 2/3/4。

给 Codex 的总任务

```text
Build a Figure2-Figure3 data-efficiency and readiness pipeline.

Scientific objective:
Determine whether the current simulation libraries for Figure 2 E_app(x,y) and Figure 3
E(x,y,z) are sufficient for posterior inference, not by sample count, but by:
1. learning-curve saturation,
2. simulator-support coverage of real MMP phantom / animal / ex vivo readouts,
3. posterior calibration and sharpness,
4. OOD/failure utility.

The pipeline must compare:
- U-Net / direct regression,
- vanilla diffusion,
- DPS-style inverse diffusion where available,
- operator-conditioned diffusion (ours).

The result should produce:
- quantitative metrics,
- PASS/YELLOW/FAIL gates,
- Figure 2 panels,
- Figure 3 panels,
- a Figure 4 posterior-bank readiness report.

Hard rule:
Do not claim real paired piezoelectric / ultrasound experiments unless actual paired hardware
data exist. Piezo and ultrasound remain simulated benchmark operators unless real files are
present.
```

---

1. Required data layout

让 Codex 先建立如下目录结构。没有的文件先标记 `MISSING`，不要自动伪造真实数据。

```text
```

```
data/
  sim2d/
    library_Eapp_2d.npz
    simulator_params_2d.csv
    readouts_mmp_2d.npz
    readouts_piezo_2d.npz
    readouts_us_2d.npz

  sim3d/
    library_E_xyz_3d.npz
    simulator_params_3d.csv
    readouts_mmp_3d.npz
    readouts_piezo_3d.npz
    readouts_us_3d.npz
    depth_truth.csv

  real_mmp/
    phantom/
      readouts/*.npz
      metadata.csv
      reference_maps/*.npz
    animal/
      readouts/*.npz
      metadata.csv
      optional_reference/*.npz
    ex_vivo/
      readouts/*.npz
      metadata.csv
      optional_histology_afm_reference/*.npz

  splits/
    figure2_config_splits.json
    figure3_config_splits.json

  posterior_bank/
    fig2_posterior_bank.npz
    fig3_posterior_bank.npz
    fig4_ready_posterior_bank.npz

reports/
  figure23_data_inventory.md
  figure23_data_efficiency_report.md
  figure23_simulator_support_report.md
  figure23_gate_report.md
  figure4_entry_report.md
```

每个样本必须有统一 metadata：

```csv
sample_id, config_id, source_label, operator_id, split, seed,
lesion_type, morphology_group, contrast_group,
depth_group, z_top_mm, z_bottom_mm, thickness_mm,
contact_pressure, calibration_shift, noise_level,
readout_path, target_path, reference_path
```

其中：

```text
source_label ∈ {
  SIMULATED_TRAIN,
  SIMULATED_VAL,
  SIMULATED_TEST,
  REAL_MMP_PHANTOM,
  REAL_MMP_ANIMAL,
  REAL_MMP_EX_VIVO,
  MISSING
}
```

关键：**split 必须按 config_id 划分，不允许同一个 physical configuration 的重复 readout 同时出现在 train 和 test。**

---

2. Codex 脚本清单

让 Codex 实现这些脚本：

```text
scripts/
  fig23_00_inventory.py
  fig23_01_make_config_splits.py
  fig23_02_extract_readout_features.py
  fig23_03_support_coverage.py
  fig23_04_train_learning_curve_sweep.py
  fig23_05_run_posterior_inference.py
  fig23_06_evaluate_fig2_metrics.py
  fig23_07_evaluate_fig3_metrics.py
  fig23_08_fit_learning_curves.py
  fig23_09_calibration_and_conformal.py
  fig23_10_ood_failure_utility.py
  fig23_11_projection_consistency.py
  fig23_12_gate_decision.py
  fig23_13_build_figure_panels.py
```

```
fig23_14_export_figure4_posterior_bank.py
```

输出核心文件：

```text
data/qc/fig2_metrics_by_N.csv
data/qc/fig3_metrics_by_N.csv
data/qc/support_coverage_real_mmp.csv
data/qc/learning_curve_fits.csv
data/qc/calibration_metrics.csv
data/qc/ood_failure_metrics.csv
data/qc/projection_consistency_metrics.csv
data/qc/gate_decisions.json
reports/figure23_gate_report.md
```

---

3. Gate 0: data inventory and leakage audit

Codex 首先运行：

```bash
python scripts/fig23_00_inventory.py \
  --sim2d data/sim2d \
  --sim3d data/sim3d \
  --real data/real_mmp \
  --out reports/figure23_data_inventory.md
```

必须检查：

```text
1. Figure 2 E_app(x,y) truth exists.
2. Figure 3 E(x,y,z), z_top, z_bottom, thickness truth exists.
3. Operator readouts exist for MMP / piezo / ultrasound.
4. Real MMP phantom readouts exist.
5. Real MMP animal readouts exist.
6. Real MMP ex vivo readouts exist.
7. Reference maps exist where claimed.
8. No paired piezo / ultrasound real data are claimed unless real files exist.
```

Leakage audit：

```text
```

```
FAIL if:
- same config_id appears in train and test;
- same latent E_app or E_xyz appears in train and test under different readout noise;
- normalization stats are computed using test or real readouts;
- q_d metadata contains true z_bottom, thickness, or lesion label not available at inference;
- posterior_mean is constructed from truth rather than from y_d.
```

---

4. Gate 1: simulator support coverage

这是证明真实 MMP phantom / animal / ex vivo readouts 落在模拟 manifold 内的核心。

4.1 Readout feature extraction

Codex 需要为每个 readout y_d 提取多组特征：

```text
A. Raw statistical features:
  mean, std, skew, kurtosis, min, max, dynamic range

B. Spatial features:
  center-of-mass, eccentricity, radial profile, gradient energy,
  Laplacian energy, local contrast, edge density

C. Spectral features:
  Fourier low/mid/high-frequency energy,
  anisotropy, power spectrum slope

D. Operator-physics features:
  estimated contact pressure,
  edge-effect index,
  sensor-to-sensor coupling,
  peak-to-background ratio,
  Hall-array spatial correlation length

E. Learned features:
  frozen readout encoder embedding from Figure 2/3 model
  or autoencoder latent representation
```

命令：

```bash
python scripts/fig23_02_extract_readout_features.py \
  --sim data/sim2d data/sim3d \
  --real data/real_mmp \
```

```
  --out data/qc/readout_features.parquet
```

4.2 Manifold support tests

对每个 real readout y_i^{real}，计算它到 simulated train manifold 的距离：

```text
1. kNN distance in feature space
2. Mahalanobis distance after robust covariance shrinkage
3. Local density / one-class model score
4. MMD between real group and matched simulated group
5. Two-sample classifier AUC: simulated vs real
6. Conformal support p-value
```

定义：

$$S_i = \mathbf{1}\left[d_{\mathrm{kNN}}(\phi(y_i^{real}), \mathcal{D}_{sim}) \leq q_{0.95}^{sim}\right]$$

Group-level support：

$$S_{\mathrm{group}} = \frac{1}{N_{\mathrm{group}}} \sum_i S_i$$

运行：

```bash
python scripts/fig23_03_support_coverage.py \
  --features data/qc/readout_features.parquet \
  --groups REAL_MMP_PHANTOM REAL_MMP_ANIMAL REAL_MMP_EX_VIVO \
  --out data/qc/support_coverage_real_mmp.csv \
  --figout figures/qc/support_umap.png
```

4.3 Support gate

```text
PASS:
  Overall ≥90% real MMP readouts inside 95% synthetic support envelope.
  Each group, phantom / animal / ex vivo, ≥80% inside support.
  Two-sample classifier AUC ≤0.70.
  MMD not significant after permutation test, or effect size small.

YELLOW:
  Overall 75-90% inside support, or one group <80%.
  Can show simulation-grounded results, but Figure 4 sim-to-real claim must be softened.
```

```
FAIL:
  Overall <75% inside support.
  Expand simulator or calibrate operator before final Figure 2/3.
```

注意：如果真实 ex vivo 数据很少，不能只用 p-value。要同时报告 effect size、nearest-neighbor examples 和 support-distance distribution。

Figure 中如何体现：

```text
Figure 2b:
  show UMAP/PCA of simulated MMP readouts with real phantom/animal/ex vivo points overlaid.
  Add "real-readout envelope coverage: PASS/YELLOW/FAIL".

Figure 3b or 3f:
  show support-distance stratified by depth group and source_label.
```

---

5. Gate 2: learning-curve saturation / data efficiency

5.1 Training sizes

对 Figure 2 和 Figure 3 分别跑 simulation-size sweep：

```text
N_grid_2D = [500, 1k, 2k, 5k, 10k, 25k, 50k]
N_grid_3D = [1k, 2k, 5k, 10k, 25k, 50k, 100k]
seeds = [0, 1, 2]
```

模型：

```text
Figure 2:
  bicubic / kernel interpolation
  U-Net / direct regression
  vanilla diffusion
  DPS-style inverse diffusion
  operator-conditioned diffusion

Figure 3:
  direct scalar regression
  3D U-Net
  kernel/GP depth-response baseline
```

```
vanilla 3D diffusion
DPS-style inverse diffusion
operator-conditioned 3D diffusion
```

命令：

```bash
python scripts/fig23_04_train_learning_curve_sweep.py \
  --tasks fig2 fig3 \
  --N_grid 500 1000 2000 5000 10000 25000 50000 100000 \
  --models unet vanilla_diffusion dps opcond_diffusion \
  --seeds 0 1 2 \
  --config_split data/splits \
  --out data/qc/training_runs.csv
```

如果 GPU 不足，Codex 应支持：

```text
--use_existing_checkpoints
--proxy_epochs
--frozen_encoder
--likelihood_weighted_posterior
```

CPU-only 下可以先做 likelihood-weighted library posterior，但必须标注为 `POSTERIOR_SURROGATE`
。

5.2 Learning-curve fitting

对每个 metric $m(N)$ 拟合：

lower-is-better:

$$m(N) = m_\infty + aN^{-\alpha}$$

higher-is-better:

$$m(N) = m_\infty - aN^{-\alpha}$$

保存：

```text
m_inf
alpha
N90
N95
```

```
AULC
marginal_gain_last_doubling
bootstrap_CI
seed_SD
```

定义：

```text
N90:
  simulation size needed to reach 90% of achievable improvement.

N95:
  simulation size needed to reach 95% of achievable improvement.

DER:
  data-efficiency ratio = N90(baseline) / N90(ours)
```

运行：

```bash
python scripts/fig23_08_fit_learning_curves.py \
  --metrics data/qc/fig2_metrics_by_N.csv data/qc/fig3_metrics_by_N.csv \
  --out data/qc/learning_curve_fits.csv \
  --figout figures/qc/learning_curves/
```

5.3 Learning-curve gate

```text
PASS:
  Last-doubling improvement <3-5% for primary metrics.
  Current library size ≥ estimated N90.
  Seed SD <10% of mean for primary metrics.
  Operator-conditioned diffusion has DER ≥2 for CRPS or calibrated posterior score versus U-Net/vanilla diffusion.

YELLOW:
  Metrics improving slowly but N90 reached for map error, not for calibration.
  Can proceed with reconstruction claim, not strong posterior/reliability claim.

FAIL:
  Learning curve still steep after final N.
  Current simulation library not sufficient.
```

Figure 中如何体现：

```text
Figure 2f:
  MAE(E_app) and CRPS learning curves versus N.
  Bar chart of N90 for U-Net, vanilla diffusion, op-conditioned diffusion.

Figure 3f:
  z_bottom MAE, CRPS, |Cov90−0.90| and AUROC_fail learning curves versus N.
```

---

6. Gate 3: Figure 2 metrics for $E_{\text{app}}(x, y)$

Figure 2 的评价对象是 2D apparent mechanics posterior：

$$p(E_{\text{app}}(x, y), B, C \mid y_d, d, q_d)$$

不要把 Figure 2 变成 depth problem。

6.1 Figure 2 metrics

```text
Map fidelity:
  MAE(E_app)
  RMSE(E_app)
  normalized MAE
  SSIM
  PSNR, optional

Boundary:
  Dice
  Boundary-F1
  Hausdorff95
  centroid shift
  lesion area error

Contrast:
  contrast MAE
  log contrast MAE
  peak-to-background ratio error

Posterior:
  pixelwise CRPS
  lesion-level CRPS
  Cov90(E_app)
  Width90(E_app)
  UCE/ENCE
```

```
Physics consistency:
  re-forward residual ||H_d(E_app, q_d) - y_d||
  residual spatial autocorrelation

Failure utility:
  AUROC_fail for high map error
  risk-coverage / AUSE
```

命令：

```bash
python scripts/fig23_06_evaluate_fig2_metrics.py \
  --posterior data/posterior_bank/fig2_posterior_bank.npz \
  --truth data/sim2d/library_Eapp_2d.npz \
  --out data/qc/fig2_metrics_by_N.csv
```

6.2 Figure 2 pass gate

```text
PASS:
  MAE(E_app) plateau reached.
  Boundary-F1 plateau reached.
  CRPS improves over U-Net and vanilla diffusion.
  Cov90 between 0.87 and 0.93 after calibration, or explicitly reported as calibrated ± CI.
  Width90 not inflated beyond acceptable lesion contrast resolution.
  Forward residual improves over baselines.
  Real MMP phantom readouts inside simulator support.

YELLOW:
  map quality passes, posterior calibration not yet stable.
  Use Figure 2 for apparent-map reconstruction, but move reliability claim to Figure 4.

FAIL:
  real phantom support fails or learning curve not saturated.
```

---

7. Gate 4: Figure 3 metrics for $E(x, y, z)$

Figure 3 的评价对象是 continuous depth posterior，不是 nominal depth band。你之前的 depth evaluation note 已经明确：顶层汇报要按 config-level 聚合，再对 depth-group 做 macro average，并报告 seed mean/SD，避免 repeated scans 虚高样本数。

7.1 Figure 3 primary posterior target

$$F_i(z_{\text{bottom}}) = p(z_{\text{bottom}} \mid y_d, d, q_d)$$

点估计：

$$\hat{z}_i = \text{median}(\tilde{z}_{i1:M})$$

主要指标：

```text
Point:
  MAE(z_bottom)
  RMSE(z_bottom)
  Bias(z_bottom)
  P90AE(z_bottom)
  MedAE(z_bottom)

Probabilistic:
  CRPS(z_bottom)
  WIS(z_bottom), if using quantiles only

Interval:
  MAE(z_top)
  MAE(thickness)
  interval IoU for [z_top, z_bottom]

Calibration:
  Cov50 / Cov80 / Cov90 / Cov95
  Width50 / Width80 / Width90 / Width95
  PIT
  UCE / ENCE

Multimodality:
  n_modes
  mode locations
  mode weights
  valley ratio
  multimodality frequency

OOD:
  error vs d_out slope
  uncertainty vs d_out slope
  AUROC_fail
  AUPRC_fail
  risk-coverage
  AUSE
```

```
Trend:
  extrapolation slope beta
  range compression ratio
  Spearman rho across held-out depth groups
```

命令：

```bash
python scripts/fig23_07_evaluate_fig3_metrics.py \
  --posterior data/posterior_bank/fig3_posterior_bank.npz \
  --truth data/sim3d/depth_truth.csv \
  --aggregate config_then_depth_macro \
  --out data/qc/fig3_metrics_by_N.csv
```

7.2 CRPS implementation

对 posterior samples：

$$\mathrm{CRPS}(F_i, y_i) = \frac{1}{M}\sum_{m}|\tilde{y}_{im} - y_i| - \frac{1}{2M^2}\sum_{m,n}|\tilde{y}_{im} - \tilde{y}_{in}|$$

Codex 应保存 full posterior samples；如果没有 samples，只能用 WIS。

7.3 Figure 3 pass gate

```text
PASS:
  z_bottom MAE and CRPS learning curves plateau.
  CRPS improves ≥15-25% over strongest non-operator baseline.
  Cov90 between 0.87 and 0.93 for in-support cases.
  Width90 not inflated.
  interval IoU ≥0.55-0.60.
  AUROC_fail ≥0.75-0.80.
  Multimodality appears in low-observability cases rather than being collapsed.
  Piezo deep cases show broad/multimodal posterior, not false overconfidence.

YELLOW:
  point depth error passes but calibration weak.
  Can show posterior qualitatively, but Figure 4 reliability claim must be labeled under
calibration repair.

FAIL:
  posterior collapse, Cov90 near 1.00 with very large Width90, or uncertainty unrelated to
```

```
error.
```

8. Gate 5: posterior calibration and conformal repair

Codex 应做 validation-set calibration，不允许用 test set 调参。

```bash
python scripts/fig23_09_calibration_and_conformal.py \
  --fig2 data/posterior_bank/fig2_posterior_bank.npz \
  --fig3 data/posterior_bank/fig3_posterior_bank.npz \
  --val_split data/splits/validation_config_ids.json \
  --out data/qc/calibration_metrics.csv
```

Calibration methods:

```text
1. posterior scale calibration:
   z_samples_calibrated = median + s_operator * (samples - median)

2. isotonic calibration for scalar uncertainty scores

3. conformalized quantile adjustment:
   q_low'  = q_low  - Δ_operator
   q_high' = q_high + Δ_operator

4. group-aware calibration:
   separate calibration by operator_id and depth_group if enough validation samples exist
```

Calibration metrics:

```text
Cov50, Width50
Cov80, Width80
Cov90, Width90
Cov95, Width95
|Cov90 - 0.90|
UCE
ENCE
PIT deviation
standardized residual mean/std
```

Calibration gate:

```text
PASS:
  |Cov90 - 0.90| ≤0.03 for ID / ≤0.05 for hard OOD.
  Width90 decreases or remains reasonable after calibration.
  PIT not strongly U-shaped or inverse-U.
  UCE/ENCE improves over uncalibrated model.

YELLOW:
  coverage fixed only by broadening interval too much.
  Use as conservative posterior, not strong calibrated posterior.

FAIL:
  coverage and width both poor.
```

---

9. Gate 6: OOD / failure utility

这里是 Figure 4 的核心入口。Codex 必须证明 uncertainty 不是装饰，而是能抓住高风险 reconstruction。

```bash
python scripts/fig23_10_ood_failure_utility.py \
  --fig2 data/posterior_bank/fig2_posterior_bank.npz \
  --fig3 data/posterior_bank/fig3_posterior_bank.npz \
  --support data/qc/support_coverage_real_mmp.csv \
  --out data/qc/ood_failure_metrics.csv
```

9.1 Figure 2 failure

定义：

$$f_i^{2D} = \mathbf{1}[\mathrm{MAE}(E_{\mathrm{app},i}) > \tau_{\mathrm{map}}]$$

or:

```text
boundary failure:
  Boundary-F1 < threshold

forward failure:
  re-forward residual > threshold
```

9.2 Figure 3 failure

定义：

$$f_i^{3D} = \mathbf{1}[|\hat{z}_{\mathrm{bottom},i} - z_{\mathrm{bottom},i}^{true}| > \tau_z]$$

建议：

```text
tau_z = 1 mm
tau_z_secondary = 0.5 mm and 2 mm
```

9.3 Uncertainty scores

```text
Figure 2:
  mean pixel std
  lesion-region std
  boundary entropy
  forward residual
  support distance

Figure 3:
  total_std(z_bottom)
  epistemic_std(z_bottom), if available
  posterior entropy
  multimodality score
  forward residual
  support distance
  d_out
```

指标：

```text
AUROC_fail
AUPRC_fail
FPR@95TPR
risk-coverage curve
AUSE
risk at 80% retained samples
risk reduction after rejecting top 20% uncertainty
```

Gate:

```text
PASS:
  AUROC_fail ≥0.75-0.80.
```

```
  AUPRC_fail improves over prevalence by ≥2×.
  Rejecting top 20% uncertainty reduces MAE or CRPS by ≥25%.
  AUSE improves ≥15-20% over baseline.
  uncertainty increases with support distance and d_out.

YELLOW:
  uncertainty increases with OOD, but failure AUROC weak.

FAIL:
  high uncertainty does not correspond to high error.
```

---

10. Gate 7: Figure 2–3 projection consistency

如果 Figure 2 是 $E_{\text{app}}(x,y)$，Figure 3 是 $E(x,y,z)$，两者合并时必须检查：

$$E_{\text{app}}^{3D\to 2D}(x,y) = P_z[E(x,y,z)]$$

Codex 要比较：

```text
Figure 2 posterior E_app
vs
Figure 3 posterior projection P_z[E(x,y,z)]
```

命令：

```bash
python scripts/fig23_11_projection_consistency.py \
  --fig2 data/posterior_bank/fig2_posterior_bank.npz \
  --fig3 data/posterior_bank/fig3_posterior_bank.npz \
  --out data/qc/projection_consistency_metrics.csv
```

指标：

```text
NMAE(E_app_2D, E_app_3D_projection)
SSIM
Boundary-F1
centroid shift
lesion area difference
contrast rank Spearman rho
posterior correlation
```

Gate:

```text
PASS:
  NMAE ≤0.10-0.15.
  SSIM ≥0.80-0.85.
  centroid shift ≤0.5-1.0 mm.
  area difference ≤15-20%.
  contrast-rank Spearman rho ≥0.80.

YELLOW:
  map projection consistent but boundary shifts.
  Use examples carefully.

FAIL:
  Figure 2 and Figure 3 are not reconstructing consistent biomechanical states.
```

---

11. Final Figure 4 entry gate

Figure 4 只能接收通过 readiness gate 的 posterior bank。

```bash
python scripts/fig23_14_export_figure4_posterior_bank.py \
  --fig2 data/posterior_bank/fig2_posterior_bank.npz \
  --fig3 data/posterior_bank/fig3_posterior_bank.npz \
  --qc data/qc/gate_decisions.json \
  --out data/posterior_bank/fig4_ready_posterior_bank.npz
```

必须包含：

```text
case_id
config_id
split_id
seed_id
source_label
operator_id
operator_metadata
morphology_group
depth_group
support_distance
d_out
```

```
posterior_samples_Eapp
posterior_samples_E_xyz
posterior_samples_zbottom
posterior_samples_ztop
posterior_samples_thickness

posterior_quantiles
CRPS
WIS
PIT
Cov50/Cov80/Cov90/Cov95 flags
Width50/Width80/Width90/Width95
UCE/ENCE bin id

uncertainty_total
uncertainty_epistemic
uncertainty_aleatoric
forward_residual
multimodality metrics

failure_label_tau_map
failure_label_tau_0p5mm
failure_label_tau_1mm
failure_label_tau_2mm
risk_coverage_inputs
task_utility_inputs
```

Figure 4 entry decision:

```text
PASS:
  Gate 1 support coverage PASS
  Gate 2 learning plateau PASS
  Gate 3 Figure 2 reconstruction PASS
  Gate 4 Figure 3 posterior PASS
  Gate 5 calibration PASS
  Gate 6 OOD utility PASS
  Gate 7 projection consistency PASS
  posterior bank complete

YELLOW:
  one gate is YELLOW, none FAIL.
  Figure 4 can include reliability limitation / calibration repair.

FAIL:
  any hard leakage, support, calibration, or posterior collapse failure.
```

12. Figure 2 里面如何体现

Figure 2 应该展示 **2D apparent mechanics library is sufficient**。

建议 panel 安排：

```text
Figure 2b:
  2D simulation library coverage:
    lesion size, irregularity, log contrast, boundary roughness, operator noise.
  Add support checklist:
    morphology coverage PASS
    operator variation PASS
    real MMP phantom envelope PASS

Figure 2c:
  MMP / piezo / US 2D operator observability kernels.

Figure 2e:
  representative E_app reconstructions:
    readout, truth, U-Net, vanilla diffusion, ours, uncertainty, residual.

Figure 2f:
  four compact metric modules:
    1. MAE(E_app), Boundary-F1, SSIM, CRPS
    2. Cov90/Width90 or |Cov90−0.90| + Width90
    3. learning curves: MAE and CRPS versus N
    4. real-readout support distance / failure AUROC
```

Figure 2 conclusion:

```text
Domain-randomized 2D apparent-mechanics simulation reaches learning-curve saturation and covers
real MMP phantom readouts, enabling calibrated E_app posterior reconstruction.
```

不要写：

```text
The model creates better stiffness images.
```

13. Figure 3 里面如何体现

Figure 3 应该展示 **3D latent mechanics posterior is ready for reliability analysis**。

建议 panel 安排：

```text
Figure 3b:
  3D prior support:
    z_bottom, thickness, log contrast, layer variation, operator/contact variation.
  Add depth-stratified balance and real-readout envelope checklist.

Figure 3c:
  operator depth observability:
    MMP near-intermediate, piezo surface-biased, US broader/deeper.

Figure 3e:
  posterior gallery:
    shallow, mid, deep, low contrast, irregular, OOD.
  Show multimodal p(z_bottom) for ambiguous and piezo/deep cases.

Figure 3f:
  four compact modules:
    1. continuous posterior: MAE(z_bottom), CRPS, interval IoU, AUROC_fail
    2. calibration/sharpness: |Cov90−0.90|, Width90
    3. data efficiency: z_bottom MAE and CRPS learning curves
    4. OOD/operator observability: error vs d_out, multimodality frequency by operator
```

Figure 3 conclusion:

```text
3D operator-conditioned posterior sampling reaches data-efficiency and calibration gates,
exposes operator-dependent depth observability, and passes calibrated samples to Figure 4.
```

---

14. Codex master prompt

下面可以直接放给 Codex。

```text
You are building the Figure 2-3 data-efficiency, simulator-support, calibration and readiness
pipeline for an operator-conditioned inverse biomechanics manuscript.

Scientific framing:
Figure 2 reconstructs 2D apparent mechanics E_app(x,y).
Figure 3 reconstructs latent 3D mechanics E(x,y,z) and continuous depth posteriors.
The goal is not to show prettier generated maps. The goal is to demonstrate that diffusion is a
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posterior sampler for noisy nonlinear inverse problems, and that the simulation library is
sufficient by objective QC gates:
1. learning-curve saturation,
2. simulator-support coverage of real MMP phantom / animal / ex vivo readouts,
3. posterior calibration and sharpness,
4. OOD/failure utility.

Do not fabricate real data.
Do not claim real paired piezoelectric or ultrasound experiments unless corresponding files
exist.
All piezo and ultrasound results are SIMULATED_BENCHMARK unless real paired data are present.

Implement these scripts:
- scripts/fig23_00_inventory.py
- scripts/fig23_01_make_config_splits.py
- scripts/fig23_02_extract_readout_features.py
- scripts/fig23_03_support_coverage.py
- scripts/fig23_04_train_learning_curve_sweep.py
- scripts/fig23_05_run_posterior_inference.py
- scripts/fig23_06_evaluate_fig2_metrics.py
- scripts/fig23_07_evaluate_fig3_metrics.py
- scripts/fig23_08_fit_learning_curves.py
- scripts/fig23_09_calibration_and_conformal.py
- scripts/fig23_10_ood_failure_utility.py
- scripts/fig23_11_projection_consistency.py
- scripts/fig23_12_gate_decision.py
- scripts/fig23_13_build_figure_panels.py
- scripts/fig23_14_export_figure4_posterior_bank.py

Data splits:
Use config-level splitting, not scan-level splitting.
No repeated readout or same latent configuration may appear in both train and test.
Compute normalization only from train split.

Models:
Figure 2:
- interpolation / kernel baseline
- U-Net or direct regression
- vanilla diffusion
- DPS-style inverse diffusion if available
- operator-conditioned diffusion

Figure 3:
- direct scalar regression
- 3D U-Net
- kernel/GP depth-response baseline
- vanilla 3D diffusion
- DPS-style inverse diffusion if available
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- operator-conditioned 3D diffusion

Training sweep:
N_grid_2D = [500, 1000, 2000, 5000, 10000, 25000, 50000]
N_grid_3D = [1000, 2000, 5000, 10000, 25000, 50000, 100000]
seeds = [0, 1, 2]

Learning curve:
Fit m(N) = m_inf + a*N^(-alpha) for lower-is-better metrics.
Fit m(N) = m_inf - a*N^(-alpha) for higher-is-better metrics.
Report N90, N95, AULC, marginal gain after last doubling, bootstrap CI and seed SD.
Report data-efficiency ratio = N90(baseline) / N90(ours).

Simulator support:
Extract readout features from simulated and real MMP readouts:
raw statistics, spatial features, spectral features, operator-physics features, frozen readout
encoder embeddings.
Compute kNN distance, Mahalanobis distance, local density, MMD, two-sample classifier AUC and
conformal support p-values.
Real MMP groups:
REAL_MMP_PHANTOM, REAL_MMP_ANIMAL, REAL_MMP_EX_VIVO.
PASS support if overall ≥90% real readouts are inside the 95% synthetic support envelope and
every group is ≥80%.

Figure 2 metrics:
MAE(E_app), RMSE(E_app), NMAE(E_app), SSIM, Dice, Boundary-F1, Hausdorff95, centroid shift,
contrast error, pixelwise CRPS, Cov90, Width90, UCE/ENCE, re-forward residual, AUROC_fail,
AUSE.

Figure 3 metrics:
MAE(z_bottom), RMSE(z_bottom), Bias(z_bottom), P90AE, MedAE, CRPS(z_bottom), WIS,
Cov50/80/90/95, Width50/80/90/95, PIT, UCE/ENCE, interval IoU for [z_top,z_bottom],
MAE(thickness), multimodality frequency, AUROC_fail, AUPRC_fail, risk-coverage, AUSE, error-vs-
d_out slope, uncertainty-vs-d_out slope, extrapolation beta, range compression ratio, Spearman
rho.

Calibration:
Use validation set only.
Apply posterior scale calibration, conformal quantile adjustment and optional isotonic
calibration.
Do not fix coverage by making intervals excessively broad.
Report |Cov90 - 0.90| together with Width90.

Projection consistency:
Compare Figure 2 E_app posterior against projection of Figure 3 E(x,y,z).
PASS if NMAE ≤0.10-0.15, SSIM ≥0.80-0.85, centroid shift ≤0.5-1.0 mm, area difference ≤15-
20%, Spearman contrast rank ≥0.80.
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Gate decisions:
PASS / YELLOW / FAIL for:
- data inventory and leakage
- simulator support
- learning plateau
- Figure 2 reconstruction
- Figure 3 posterior
- calibration
- OOD/failure utility
- projection consistency
- Figure 4 posterior-bank completeness

Figure outputs:
Create figure-ready panels:
- Figure 2: support UMAP, 2D learning curves, 2D metrics, calibration/sharpness, real phantom support, representative E_app reconstructions.
- Figure 3: 3D learning curves, continuous depth posterior metrics, calibration/sharpness, OOD/failure, multimodality, operator observability.
- Figure 4: posterior bank summary and readiness report.

Final files:
- data/qc/fig2_metrics_by_N.csv
- data/qc/fig3_metrics_by_N.csv
- data/qc/support_coverage_real_mmp.csv
- data/qc/learning_curve_fits.csv
- data/qc/calibration_metrics.csv
- data/qc/ood_failure_metrics.csv
- data/qc/projection_consistency_metrics.csv
- data/qc/gate_decisions.json
- reports/figure23_gate_report.md
- reports/figure4_entry_report.md
- data/posterior_bank/fig4_ready_posterior_bank.npz

Hard fail:
- posterior mean suspiciously equals truth;
- test leakage by config_id;
- real readouts outside support but reported as validated;
- Cov90 close to 1.00 only because Width90 is huge;
- uncertainty does not predict failure;
- Figure 2 and Figure 3 projections are inconsistent.
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15. Manuscript wording after this pipeline passes

可以写：

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We assessed simulation sufficiency using a pre-specified data-efficiency and readiness protocol
rather than by the number of simulated samples. Learning curves were fitted across
progressively larger domain-randomized libraries, real MMP phantom/animal/ex vivo readouts were
tested against the simulated readout manifold, and posterior calibration and failure detection
were evaluated before passing samples to the Figure 4 reliability analysis.
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Diffusion 的表述：

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The diffusion model is used as an operator-conditioned posterior sampler for a noisy nonlinear
inverse problem, not as an image generator. Its advantage is evaluated by CRPS, calibration,
forward consistency and failure detection under operator shift, rather than by visual sharpness
alone.
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这样 Figure 2/3 的 story 会从“simulation library 很大 + 图很好看”升级为：

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simulation support is sufficient,
training is data-efficient,
posterior is calibrated,
uncertainty is useful,
and the posterior bank is ready for Figure 4.
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